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An Elementary Course in Synthetic Projective Geometry

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PREFACE

The following course is intended to give, in as simple a way as possible, the essentials of synthetic projective geometry. While, in the main, the theory is developed along the well-beaten track laid out by the great masters of the subject, it is believed that there has been a slight smoothing of the road in some places. Especially will this be observed in the chapter on Involution. The author has never felt satisfied with the usual treatment of that subject by means of circles and anharmonic ratios. A purely projective notion ought not to be based on metrical foundations. Metrical developments should be made there, as elsewhere in the theory, by the introduction of infinitely distant elements.

The author has departed from the century-old custom of writing in parallel columns each theorem and its dual. He has not found that it conduces to sharpness of vision to try to focus his eyes on two things at once. Those who prefer the usual method of procedure can, of course, develop the two sets of theorems side by side; the author has not found this the better plan in actual teaching.

As regards nomenclature, the author has followed the lead of the earlier writers in English, and has called the system of lines in a plane which all pass through a point a _pencil of rays_ instead of a _bundle of rays_, as later writers seem inclined to do. For a point considered as made up of all the lines and planes through it he has ventured to use the term _point system_, as being the natural dualization of the usual term _plane system_. He has also rejected the term _foci of an involution_, and has not used the customary terms for classifying involutions—_hyperbolic involution_, _elliptic involution_ and _parabolic involution_. He has found that all these terms are very confusing to the student, who inevitably tries to connect them in some way with the conic sections.

Enough examples have been provided to give the student a clear grasp of the theory. Many are of sufficient generality to serve as a basis for individual investigation on the part of the student. Thus, the third example at the end of the first chapter will be found to be very fruitful in interesting results. A correspondence is there indicated between lines in space and circles through a fixed point in space. If the student will trace a few of the consequences of that correspondence, and determine what configurations of circles correspond to intersecting lines, to lines in a plane, to lines of a plane pencil, to lines cutting three skew lines, etc., he will have acquired no little practice in picturing to himself figures in space.

The writer has not followed the usual practice of inserting historical notes at the foot of the page, and has tried instead, in the last chapter, to give a consecutive account of the history of pure geometry, or, at least, of as much of it as the student will be able to appreciate who has mastered the course as given in the preceding

chapters. One is not apt to get a very wide view of the history of a subject by reading a hundred biographical footnotes, arranged in no sort of sequence. The writer, moreover, feels that the proper time to learn the history of a subject is after the student has some general ideas of the subject itself.

The course is not intended to furnish an illustration of how a subject may be developed, from the smallest possible number of fundamental assumptions. The author is aware of the importance of work of this sort, but he does not believe it is possible at the present time to write a book along such lines which shall be of much use for elementary students. For the purposes of this course the student should have a thorough grounding in ordinary elementary geometry so far as to include the study of the circle and of similar triangles. No solid geometry is needed beyond the little used in the proof of Desargues' theorem (25), and, except in certain metrical developments of the general theory, there will be no call for a knowledge of trigonometry or analytical geometry. Naturally the student who is equipped with these subjects as well as with the calculus will be a little more mature, and may be expected to follow the course all the more easily. The author has had no difficulty, however, in presenting it to students in the freshman class at the University of California.

The subject of synthetic projective geometry is, in the opinion of the writer, destined shortly to force its way down into the secondary schools; and if this little book helps to accelerate the movement, he will feel amply repaid for the task of working the materials into a form available for such schools as well as for the lower classes in the university.

The material for the course has been drawn from many sources. The author is chiefly indebted to the classical works of Reye, Cremona, Steiner, Poncelet, and Von Staudt. Acknowledgments and thanks are also due to Professor Walter C. Eells, of the U.S. Naval Academy at Annapolis, for his searching examination and keen criticism of the manuscript; also to Professor Herbert Ellsworth Slaught, of The University of Chicago, for his many valuable suggestions, and to Professor B. M. Woods and Dr. H. N. Wright, of the University of California, who have tried out the methods of presentation, in their own classes.

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CHAPTER I - ONE-TO-ONE CORRESPONDENCE

1. Definition of one-to-one correspondence. Given any two sets of individuals, if it is possible to set up such a correspondence between the two sets that to any individual in one set corresponds one and only one individual in the other, then the two sets are said to be in _one-to-one correspondence_ with each other. This notion, simple as it is,

is of fundamental importance in all branches of science. The process of counting is nothing but a setting up of a one-to-one correspondence between the objects to be counted and certain words, ’one,’ ’two,’ ’three,’ etc., in the mind. Many savage peoples have discovered no better method of counting than by setting up a one-to-one correspondence between the objects to be counted and their fingers. The scientist who busies himself with naming and classifying the objects of nature is only setting up a one-to-one correspondence between the objects and certain words which serve, not as a means of counting the objects, but of listing them in a convenient way. Thus he may be able to marshal and array his material in such a way as to bring to light relations that may exist between the objects themselves. Indeed, the whole notion of language springs from this idea of one-to-one correspondence.

2. Consequences of one-to-one correspondence. The most useful and interesting problem that may arise in connection with any one-to-one correspondence is to determine just what relations existing between the individuals of one assemblage may be carried over to another assemblage in one-to-one correspondence with it. It is a favorite error to assume that whatever holds for one set must also hold for the other. Magicians are apt to assign magic properties to many of the words and symbols which they are in the habit of using, and scientists are constantly confusing objective things with the subjective formulas for them. After the physicist has set up correspondences between physical facts and mathematical formulas, the "interpretation" of these formulas is his most important and difficult task.

3. In mathematics, effort is constantly being made to set up one-to-one correspondences between simple notions and more complicated ones, or between the well-explored fields of research and fields less known. Thus, by means of the mechanism employed in analytic geometry, algebraic theorems are made to yield geometric ones, and vice versa. In geometry we get at the properties of the conic sections by means of the properties of the straight line, and cubic surfaces are studied by means of the plane.

[Figure 1]

FIG. 1

[Figure 2]

FIG. 2

4. One-to-one correspondence and enumeration. If a one-to-one correspondence has been set up between the objects of one set and the objects of another set, then the inference may usually be drawn that they have the same number of elements. If, however, there is an infinite number of individuals in each of the two sets, the notion of counting is necessarily ruled out. It may be possible, nevertheless, to set up a one-to-one correspondence between the elements of two sets even when the number is infinite. Thus, it is easy to set up such a correspondence between the points of a line an inch long and the points of a line two inches long. For let the lines (Fig. 1) be

_AB_ and _A’B’_. Join _AA’_ and _BB’_, and let these joining lines meet in _S_. For every point _C_ on _AB_ a point _C’_ may be found on _A’B’_ by joining _C_ to _S_ and noting the point _C’_ where _CS_ meets _A’B’_. Similarly, a point _C_ may be found on _AB_ for any point _C’_ on _A’B’_. The correspondence is clearly one-to-one, but it would be absurd to infer from this that there were just as many points on _AB_ as on _A’B’_. In fact, it would be just as reasonable to infer that there were twice as many points on _A’B’_ as on _AB_. For if we bend _A’B’_ into a circle with center at _S_ (Fig. 2), we see that for every point _C_ on _AB_ there are two points on _A’B’_. Thus it is seen that the notion of one-to-one correspondence is more extensive than the notion of counting, and includes the notion of counting only when applied to finite assemblages.

5. Correspondence between a part and the whole of an infinite assemblage. In the discussion of the last paragraph the remarkable fact was brought to light that it is sometimes possible to set the elements of an assemblage into one-to-one correspondence with a part of those elements. A moment’s reflection will convince one that this is never possible when there is a finite number of elements in the assemblage.—Indeed, we may take this property as our definition of an infinite assemblage, and say that an infinite assemblage is one that may be put into one-to-one correspondence with part of itself. This has the advantage of being a positive definition, as opposed to the usual negative definition of an infinite assemblage as one that cannot be counted.

6. Infinitely distant point. We have illustrated above a simple method of setting the points of two lines into one-to-one correspondence. The same illustration will serve also to show how it is possible to set the points on a line into one-to-one correspondence with the lines through a point. Thus, for any point _C_ on the line _AB_ there is a line _SC_ through _S_. We must assume the line _AB_ extended indefinitely in both directions, however, if we are to have a point on it for every line through _S_; and even with this extension there is one line through _S_, according to Euclid’s postulate, which does not meet the line _AB_ and which therefore has no point on _AB_ to correspond to it. In order to smooth out this discrepancy we are accustomed to assume the existence of an _infinitely distant_ point on the line _AB_ and to assign this point as the corresponding point of the exceptional line of _S_. With this understanding, then, we may say that we have set the lines through a point and the points on a line into one-to-one correspondence. This correspondence is of such fundamental importance in the study of projective geometry that a special name is given to it. Calling the totality of points on a line a _point-row_, and the totality of lines through a point a _pencil of rays_, we say that the point-row and the pencil related as above are in _perspective position_, or that they are _perspectively related_.

7. Axial pencil; fundamental forms. A similar correspondence may be set up between the points on a line and the planes through another line which does not meet the first. Such a system of planes is called an _axial pencil_, and the three assemblages—the point-row, the pencil of rays, and the axial pencil—are called _fundamental forms_. The fact that they may all be set into one-to-one

correspondence with each other is expressed by saying that they are of the same order. It is usual also to speak of them as of the first order. We shall see presently that there are other assemblages which cannot be put into this sort of one-to-one correspondence with the points on a line, and that they will very reasonably be said to be of a higher order.

8. Perspective position. We have said that a point-row and a pencil of rays are in perspective position if each ray of the pencil goes through the point of the point-row which corresponds to it. Two pencils of rays are also said to be in perspective position if corresponding rays meet on a straight line which is called the axis of perspectivity. Also, two point-rows are said to be in perspective position if corresponding points lie on straight lines through a point which is called the center of perspectivity. A point-row and an axial pencil are in perspective position if each plane of the pencil goes through the point on the point-row which corresponds to it, and an axial pencil and a pencil of rays are in perspective position if each ray lies in the plane which corresponds to it; and, finally, two axial pencils are perspectively related if corresponding planes meet in a plane.

9. Projective relation. It is easy to imagine a more general correspondence between the points of two point-rows than the one just described. If we take two perspective pencils, _A_ and _S_, then a point-row _a_ perspective to _A_ will be in one-to-one correspondence with a point-row _b_ perspective to _B_, but corresponding points will not, in general, lie on lines which all pass through a point. Two such point-rows are said to be _projectively related_, or simply projective to each other. Similarly, two pencils of rays, or of planes, are projectively related to each other if they are perspective to two perspective point-rows. This idea will be generalized later on. It is important to note that between the elements of two projective fundamental forms there is a one-to-one correspondence, and also that this correspondence is in general _continuous_; that is, by taking two elements of one form sufficiently close to each other, the two corresponding elements in the other form may be made to approach each other arbitrarily close. In the case of point-rows this continuity is subject to exception in the neighborhood of the point "at infinity."

10. Infinity-to-one correspondence. It might be inferred that any infinite assemblage could be put into one-to-one correspondence with any other. Such is not the case, however, if the correspondence is to be continuous, between the points on a line and the points on a plane. Consider two lines which lie in different planes, and take _m_ points on one and _n_ points on the other. The number of lines joining the _m_ points of one to the _n_ points jof the other is clearly _mn_. If we symbolize the totality of points on a line by [infinity], then a reasonable symbol for the totality of lines drawn to cut two lines would be [infinity]2. Clearly, for every point on one line there are [infinity] lines cutting across the other, so that the correspondence might be called [infinity]-to-one. Thus the assemblage of lines cutting across two lines is of higher order than the assemblage of points on a line; and as we have called the point-row an assemblage of the first order, the system of lines cutting across two lines ought to be called of the second order.

11. Infinitudes of different orders. Now it is easy to set up a one-to-one correspondence between the points in a plane and the system of lines cutting across two lines which lie in different planes. In fact, each line of the system of lines meets the plane in one point, and each point in the plane determines one and only one line cutting across the two given lines—namely, the line of intersection of the two planes determined by the given point with each of the given lines. The assemblage of points in the plane is thus of the same order as that of the lines cutting across two lines which lie in different planes, and ought therefore to be spoken of as of the second order. We express all these results as follows:

12. If the infinitude of points on a line is taken as the infinitude of the first order, then the infinitude of lines in a pencil of rays and the infinitude of planes in an axial pencil are also of the first order, while the infinitude of lines cutting across two "skew" lines, as well as the infinitude of points in a plane, are of the second order.

13. If we join each of the points of a plane to a point not in that plane, we set up a one-to-one correspondence between the points in a plane and the lines through a point in space. _Thus the infinitude of lines through a point in space is of the second order._

14. If to each line through a point in space we make correspond that plane at right angles to it and passing through the same point, we see that _the infinitude of planes through a point in space is of the second order._

15. If to each plane through a point in space we make correspond the line in which it intersects a given plane, we see that _the infinitude of lines in a plane is of the second order._ This may also be seen by setting up a one-to-one correspondence between the points on a plane and the lines of that plane. Thus, take a point _S_ not in the plane. Join any point _M_ of the plane to _S_. Through _S_ draw a plane at right angles to _MS_. This meets the given plane in a line _m_ which may be taken as corresponding to the point _M_. Another very important method of setting up a one-to-one correspondence between lines and points in a plane will be given later, and many weighty consequences will be derived from it.

16. Plane system and point system. The plane, considered as made up of the points and lines in it, is called a _plane system_ and is a fundamental form of the second order. The point, considered as made up of all the lines and planes passing through it, is called a _point system_ and is also a fundamental form of the second order.

17. If now we take three lines in space all lying in different planes, and select _l_ points on the first, _m_ points on the second, and _n_ points on the third, then the total number of planes passing through one of the selected points on each line will be _lmn_. It is reasonable, therefore, to symbolize the totality of planes that are determined by the [infinity] points on each of the three lines by [infinity]3, and to call it an infinitude of the _third_ order. But it is easily seen that every plane in space is included in this totality, so that _the totality of planes in space is an infinitude of the third order._

18. Consider now the planes perpendicular to these three lines. Every set of three planes so drawn will determine a point in space, and, conversely, through every point in space may be drawn one and only one set of three planes at right angles to the three given lines. It follows, therefore, that _the totality of points in space is an infinitude of the third order._

19. Space system. Space of three dimensions, considered as made up of all its planes and points, is then a fundamental form of the _third_ order, which we shall call a _space system._

20. Lines in space. If we join the twofold infinity of points in one plane with the twofold infinity of points in another plane, we get a totality of lines of space which is of the fourth order of infinity. _The totality of lines in space gives, then, a fundamental form of the fourth order._

21. Correspondence between points and numbers. In the theory of analytic geometry a one-to-one correspondence is assumed to exist between points on a line and numbers. In order to justify this assumption a very extended definition of number must be made use of. A one-to-one correspondence is then set up between points in the plane and pairs of numbers, and also between points in space and sets of three numbers. A single constant will serve to define the position of a point on a line; two, a point in the plane; three, a point in space; etc. In the same theory a one-to-one correspondence is set up between loci in the plane and equations in two variables; between surfaces in space and equations in three variables; etc. The equation of a line in a plane involves two constants, either of which may take an infinite number of values. From this it follows that there is an infinity of lines in the plane which is of the second order if the infinity of points on a line is assumed to be of the first. In the same way a circle is determined by three conditions; a sphere by four; etc. We might then expect to be able to set up a one-to-one correspondence between circles in a plane and points, or planes in space, or between spheres and lines in space. Such, indeed, is the case, and it is often possible to infer theorems concerning spheres from theorems concerning lines, and vice versa. It is possibilities such as these that, give to the theory of one-to-one correspondence its great importance for the mathematician. It must not be forgotten, however, that we are considering only _continuous_ correspondences. It is perfectly possible to set, up a one-to-one correspondence between the points of a line and the points of a plane, or, indeed, between the points of a line and the points of a space of any finite number of dimensions, if the correspondence is not restricted to be continuous.

22. Elements at infinity. A final word is necessary in order to explain a phrase which is in constant use in the study of projective geometry. We have spoken of the "point at infinity" on a straight line—a fictitious point only used to bridge over the exceptional case when we are setting up a one-to-one correspondence between the points of a line and the lines through a point. We speak of it as "a point" and not as "points," because in the geometry studied by Euclid we assume only one line through a point parallel to a given line. In the same sense we speak of all the points at infinity in a plane as lying on a line, "the line at infinity," because the straight line is the simplest locus we can imagine which has only one point in common with any line in

the plane. Likewise we speak of the "plane at infinity," because that seems the most convenient way of imagining the points at infinity in space. It must not be inferred that these conceptions have any essential connection with physical facts, or that other means of picturing to ourselves the infinitely distant configurations are not possible. In other branches of mathematics, notably in the theory of functions of a complex variable, quite different assumptions are made and quite different conceptions of the elements at infinity are used. As we can know nothing experimentally about such things, we are at liberty to make any assumptions we please, so long as they are consistent and serve some useful purpose.

PROBLEMS

1. Since there is a threefold infinity of points in space, there must be a sixfold infinity of pairs of points in space. Each pair of points determines a line. Why, then, is there not a sixfold infinity of lines in space?

2. If there is a fourfold infinity of lines in space, why is it that there is not a fourfold infinity of planes through a point, seeing that each line in space determines a plane through that point?

3. Show that there is a fourfold infinity of circles in space that pass through a fixed point. (Set up a one-to-one correspondence between the axes of the circles and lines in space.)

4. Find the order of infinity of all the lines of space that cut across a given line; across two given lines; across three given lines; across four given lines.

5. Find the order of infinity of all the spheres in space that pass through a given point; through two given points; through three given points; through four given points.

6. Find the order of infinity of all the circles on a sphere; of all the circles on a sphere that pass through a fixed point; through two fixed points; through three fixed points; of all the circles in space; of all the circles that cut across a given line.

7. Find the order of infinity of all lines tangent to a sphere; of all planes tangent to a sphere; of lines and planes tangent to a sphere and passing through a fixed point.

8. Set up a one-to-one correspondence between the series of numbers _1_, _2_, _3_, _4_, ... and the series of even numbers _2_, _4_, _6_, _8_ .... Are we justified in saying that there are just as many even numbers as there are numbers altogether?

9. Is the axiom "The whole is greater than one of its parts" applicable to infinite assemblages?

10. Make out a classified list of all the infinitudes of the first, second, third, and fourth orders mentioned in this chapter.

CHAPTER II - RELATIONS BETWEEN FUNDAMENTAL FORMS IN ONE-TO-ONE CORRESPONDENCE WITH EACH OTHER

23. Seven fundamental forms. In the preceding chapter we have called attention to seven fundamental forms: the point-row, the pencil of rays, the axial pencil, the plane system, the point system, the space system, and the system of lines in space. These fundamental forms are the material which we intend to use in building up a general theory which will be found to include ordinary geometry as a special case. We shall be concerned, not with measurement of angles and areas or line segments as in the study of Euclid, but in combining and comparing these fundamental forms and in "generating" new forms by means of them. In problems of construction we shall make no use of measurement, either of angles or of segments, and except in certain special applications of the general theory we shall not find it necessary to require more of ourselves than the ability to draw the line joining two points, or to find the point of intersections of two lines, or the line of intersection of two planes, or, in general, the common elements of two fundamental forms.

24. Projective properties. Our chief interest in this chapter will be the discovery of relations between the elements of one form which hold between the corresponding elements of any other form in one-to-one correspondence with it. We have already called attention to the danger of assuming that whatever relations hold between the elements of one assemblage must also hold between the corresponding elements of any assemblage in one-to-one correspondence with it. This false assumption is the basis of the so-called "proof by analogy" so much in vogue among speculative theorists. When it appears that certain relations existing between the points of a given point-row do not necessitate the same relations between the corresponding elements of another in one-to-one correspondence with it, we should view with suspicion any application of the "proof by analogy" in realms of thought where accurate judgments are not so easily made. For example, if in a given point-row _u_ three points, _A_, _B_, and _C_, are taken such that _B_ is the middle point of the segment _AC_, it does not follow that the three points _A’_, _B’_, _C’_ in a point-row perspective to _u_ will be so related. Relations between the elements of any form which do go over unaltered to the corresponding elements of a form projectively related to it are called _projective relations._ Relations involving measurement of lines or of angles are not projective.

25. Desargues’s theorem. We consider first the following beautiful theorem, due to Desargues and called by his name.

_If two triangles, __A__, __B__, __C__ and __A’__, __B’__, __C’__, are so situated that the lines __AA’__, __BB’__, and __CC’__ all meet in a point, then the pairs of sides __AB__ and __A’B’__, __BC__ and __B’C’__, __CA__ and __C’A’__ all meet on a straight line, and conversely._

[Figure 3]

FIG. 3

Let the lines _AA’_, _BB’_, and _CC’_ meet in the point _M_ (Fig. 3). Conceive of the figure as in space, so that _M_ is the vertex of a trihedral angle of which the given triangles are plane sections. The lines _AB_ and _A’B’_ are in the same plane and

must meet when produced, their point of intersection being clearly a point in the plane of each triangle and therefore in the line of intersection of these two planes. Call this point _P_. By similar reasoning the point _Q_ of intersection of the lines _BC_ and _B'C'_ must lie on this same line as well as the point _R_ of intersection of _CA_ and _C'A'_. Therefore the points _P_, _Q_, and _R_ all lie on the same line _m_. If now we consider the figure a plane figure, the points _P_, _Q_, and _R_ still all lie on a straight line, which proves the theorem. The converse is established in the same manner.

26. Fundamental theorem concerning two complete quadrangles. This theorem throws into our hands the following fundamental theorem concerning two complete quadrangles, a _complete quadrangle_ being defined as the figure obtained by joining any four given points by straight lines in the six possible ways.

_Given two complete quadrangles, __K__, __L__, __M__, __N__ and __K'__, __L'__, __M'__, __N'__, so related that __KL__, __K'L'__, __MN__, __M'N'__ all meet in a point __A__; __LM__, __L'M'__, __NK__, __N'K'__ all meet in a __ point __Q__; and __LN__, __L'N'__ meet in a point __B__ on the line __AC__; then the lines __KM__ and __K'M'__ also meet in a point __D__ on the line __AC__._

[Figure 4]

FIG. 4

For, by the converse of the last theorem, _KK'_, _LL'_, and _NN'_ all meet in a point _S_ (Fig. 4). Also _LL'_, _MM'_, and _NN'_ meet in a point, and therefore in the same point _S_. Thus _KK'_, _LL'_, and _MM'_ meet in a point, and so, by Desargues's theorem itself, _A_, _B_, and _D_ are on a straight line.

27. Importance of the theorem. The importance of this theorem lies in the fact that, _A_, _B_, and _C_ being given, an indefinite number of quadrangles _K'_, _L'_, _M'_, _N'_ my be found such that _K'L'_ and _M'N'_ meet in _A_, _K'N'_ and _L'M'_ in _C_, with _L'N'_ passing through _B_. Indeed, the lines _AK'_ and _AM'_ may be drawn arbitrarily through _A_, and any line through _B_ may be used to determine _L'_ and _N'_. By joining these two points to _C_ the points _K'_ and _M'_ are determined. Then the line joining _K'_ and _M'_, found in this way, must pass through the point _D_ already determined by the quadrangle _K_, _L_, _M_, _N_. _The three points __A__, __B__, __C__, given in order, serve thus to determine a fourth point __D__._

28. In a complete quadrangle the line joining any two points is called the _opposite side_ to the line joining the other two points. The result of the preceding paragraph may then be stated as follows:

Given three points, _A_, _B_, _C_, in a straight line, if a pair of opposite sides of a complete quadrangle pass through _A_, and another pair through _C_, and one of the remaining two sides goes through _B_, then the other of the remaining two sides will go through a fixed point which does not depend on the quadrangle employed.

29. Four harmonic points. Four points, _A_, _B_, _C_, _D_, related as in the preceding theorem are called _four harmonic points_. The point _D_ is called the _fourth harmonic of __B__ with respect to __A__ and __C_. Since _B_ and _D_ play exactly the same rôle in the above construction, _B__ is also the fourth harmonic of __D__ with respect to __A__ and __C_. _B_ and _D_ are called _harmonic conjugates with respect to __A__ and __C_. We proceed to show that _A_ and _C_ are also harmonic conjugates with respect to _B_ and _D_—that is, that it is possible to find a quadrangle of which two opposite sides shall pass through _B_, two through _D_, and of the remaining pair, one through _A_ and the other through _C_.

[Figure 5]

FIG. 5

Let _O_ be the intersection of _KM_ and _LN_ (Fig. 5). Join _O_ to _A_ and _C_. The joining lines cut out on the sides of the quadrangle four points, _P_, _Q_, _R_, _S_. Consider the quadrangle _P_, _K_, _Q_, _O_. One pair of opposite sides passes through _A_, one through _C_, and one remaining side through _D_; therefore the other remaining side must pass through _B_. Similarly, _RS_ passes through _B_ and _PS_ and _QR_ pass through _D_. The quadrangle _P_, _Q_, _R_, _S_ therefore has two opposite sides through _B_, two through _D_, and the remaining pair through _A_ and _C_. _A_ and _C_ are thus harmonic conjugates with respect to _B_ and _D_. We may sum up the discussion, therefore, as follows:

30. If _A_ and _C_ are harmonic conjugates with respect to _B_ and _D_, then _B_ and _D_ are harmonic conjugates with respect to _A_ and _C_.

31. Importance of the notion. The importance of the notion of four harmonic points lies in the fact that it is a relation which is carried over from four points in a point-row _u_ to the four points that correspond to them in any point-row _u'_ perspective to _u_.

To prove this statement we construct a quadrangle _K_, _L_, _M_, _N_ such that _KL_ and _MN_ pass through _A_, _KN_ and _LM_ through _C_, _LN_ through _B_, and _KM_ through _D_. Take now any point _S_ not in the plane of the quadrangle and construct the planes determined by _S_ and all the seven lines of the figure. Cut across this set of planes by another plane not passing through _S_. This plane cuts out on the set of seven planes another quadrangle which determines four new harmonic points, _A'_, _B'_, _C'_, _D'_, on the lines joining _S_ to _A_, _B_, _C_, _D_. But _S_ may be taken as any point, since the original quadrangle may be taken in any plane through _A_, _B_, _C_, _D_; and, further, the points _A'_, _B'_, _C'_, _D'_ are the intersection of _SA_, _SB_, _SC_, _SD_ by any line. We have, then, the remarkable theorem:

32. _If any point is joined to four harmonic points, and the four lines thus obtained are cut by any fifth, the four points of intersection are again harmonic._

33. Four harmonic lines. We are now able to extend the notion of harmonic elements to pencils of rays, and indeed to axial pencils. For if we define _four

harmonic rays_ as four rays which pass through a point and which pass one through each of four harmonic points, we have the theorem

Four harmonic lines are cut by any transversal in four harmonic points.

34. Four harmonic planes. We also define _four harmonic planes_ as four planes through a line which pass one through each of four harmonic points, and we may show that

Four harmonic planes are cut by any plane not passing through their common line in four harmonic lines, and also by any line in four harmonic points.

For let the planes α, β, γ, δ, which all pass through the line _g_, pass also through the four harmonic points _A_, _B_, _C_, _D_, so that α passes through _A_, etc. Then it is clear that any plane π through _A_, _B_, _C_, _D_ will cut out four harmonic lines from the four planes, for they are lines through the intersection _P_ of _g_ with the plane π, and they pass through the given harmonic points _A_, _B_, _C_, _D_. Any other plane σ cuts _g_ in a point _S_ and cuts α, β, γ, δ in four lines that meet π in four points _A'_, _B'_, _C'_, _D'_ lying on _PA_, _PB_, _PC_, and _PD_ respectively, and are thus four harmonic hues. Further, any ray cuts α, β, γ, δ in four harmonic points, since any plane through the ray gives four harmonic lines of intersection.

35. These results may be put together as follows:

Given any two assemblages of points, rays, or planes, perspectively related to each other, four harmonic elements of one must correspond to four elements of the other which are likewise harmonic.

If, now, two forms are perspectively related to a third, any four harmonic elements of one must correspond to four harmonic elements in the other. We take this as our definition of projective correspondence, and say:

36. Definition of projectivity. _Two fundamental forms are protectively related to each other when a one-to-one correspondence exists between the elements of the two and when four harmonic elements of one correspond to four harmonic elements of the other._

[Figure 6]

FIG. 6

37. Correspondence between harmonic conjugates. Given four harmonic points, _A_, _B_, _C_, _D_; if we fix _A_ and _C_, then _B_ and _D_ vary together in a way that should be thoroughly understood. To get a clear conception of their relative motion we may fix the points _L_ and _M_ of the quadrangle _K_, _L_, _M_, _N_ (Fig. 6). Then, as _B_ describes the point-row _AC_, the point _N_ describes the point-row _AM_ perspective to it. Projecting _N_ again from _C_, we get a point-row _K_ on _AL_ perspective to the point-row _N_ and thus projective to the point-row _B_. Project the point-row _K_ from _M_ and we get a point-row _D_ on _AC_ again, which is projective to the point-row _B_. For every point _B_ we have thus one and only one point _D_, and conversely. In other words, we have set up a one-to-one correspondence between

the points of a single point-row, which is also a projective correspondence because four harmonic points _B_ correspond to four harmonic points _D_. We may note also that the correspondence is here characterized by a feature which does not always appear in projective correspondences: namely, the same process that carries one from _B_ to _D_ will carry one back from _D_ to _B_ again. This special property will receive further study in the chapter on Involution.

38. It is seen that as _B_ approaches _A_, _D_ also approaches _A_. As _B_ moves from _A_ toward _C_, _D_ moves from _A_ in the opposite direction, passing through the point at infinity on the line _AC_, and returns on the other side to meet _B_ at _C_ again. In other words, as _B_ traverses _AC_, _D_ traverses the rest of the line from _A_ to _C_ through infinity. In all positions of _B_, except at _A_ or _C_, _B_ and _D_ are separated from each other by _A_ and _C_.

39. Harmonic conjugate of the point at infinity. It is natural to inquire what position of _B_ corresponds to the infinitely distant position of _D_. We have proved (§ 27) that the particular quadrangle _K_, _L_, _M_, _N_ employed is of no consequence. We shall therefore avail ourselves of one that lends itself most readily to the solution of the problem. We choose the point _L_ so that the triangle _ALC_ is isosceles (Fig. 7). Since _D_ is supposed to be at infinity, the line _KM_ is parallel to _AC_. Therefore the triangles _KAC_ and _MAC_ are equal, and the triangle _ANC_ is also isosceles. The triangles _CNL_ and _ANL_ are therefore equal, and the line _LB_ bisects the angle _ALC_. _B_ is therefore the middle point of _AC_, and we have the theorem

_The harmonic conjugate of the middle point of __AC__ is at infinity._

[Figure 7]

FIG. 7

40. Projective theorems and metrical theorems. Linear construction. This theorem is the connecting link between the general protective theorems which we have been considering so far and the metrical theorems of ordinary geometry. Up to this point we have said nothing about measurements, either of line segments or of angles. Desargues’s theorem and the theory of harmonic elements which depends on it have nothing to do with magnitudes at all. Not until the notion of an infinitely distant point is brought in is any mention made of distances or directions. We have been able to make all of our constructions up to this point by means of the straightedge, or ungraduated ruler. A construction made with such an instrument we shall call a _linear_ construction. It requires merely that we be able to draw the line joining two points or find the point of intersection of two lines.

41. Parallels and mid-points. It might be thought that drawing a line through a given point parallel to a given line was only a special case of drawing a line joining two points. Indeed, it consists only in drawing a line through the given point and through the "infinitely distant point" on the given line. It must be remembered, however, that the expression "infinitely distant point" must not be taken literally. When we say that two parallel lines meet "at infinity," we really mean that they do not

meet at all, and the only reason for using the expression is to avoid tedious statement of exceptions and restrictions to our theorems. We ought therefore to consider the drawing of a line parallel to a given line as a different accomplishment from the drawing of the line joining two given points. It is a remarkable consequence of the last theorem that a parallel to a given line and the mid-point of a given segment are equivalent data. For the construction is reversible, and if we are given the middle point of a given segment, we can construct _linearly_ a line parallel to that segment. Thus, given that _B_ is the middle point of _AC_, we may draw any two lines through _A_, and any line through _B_ cutting them in points _N_ and _L_. Join _N_ and _L_ to _C_ and get the points _K_ and _M_ on the two lines through _A_. Then _KM_ is parallel to _AC_. _The bisection of a given segment and the drawing of a line parallel to the segment are equivalent data when linear construction is used._

42. It is not difficult to give a linear construction for the problem to divide a given segment into _n_ equal parts, given only a parallel to the segment. This is simple enough when _n_ is a power of _2_. For any other number, such as _29_, divide any segment on the line parallel to _AC_ into _32_ equal parts, by a repetition of the process just described. Take _29_ of these, and join the first to _A_ and the last to _C_. Let these joining lines meet in _S_. Join _S_ to all the other points. Other problems, of a similar sort, are given at the end of the chapter.

43. Numerical relations. Since three points, given in order, are sufficient to determine a fourth, as explained above, it ought to be possible to reproduce the process numerically in view of the one-to-one correspondence which exists between points on a line and numbers; a correspondence which, to be sure, we have not established here, but which is discussed in any treatise on the theory of point sets. We proceed to discover what relation between four numbers corresponds to the harmonic relation between four points.

[Figure 8]

FIG. 8

44. Let _A_, _B_, _C_, _D_ be four harmonic points (Fig. 8), and let _SA_, _SB_, _SC_, _SD_ be four harmonic lines. Assume a line drawn through _B_ parallel to _SD_, meeting _SA_ in _A’_ and _SC_ in _C’_. Then _A’_, _B’_, _C’_, and the infinitely distant point on _A’C’_ are four harmonic points, and therefore _B_ is the middle point of the segment _A’C’_. Then, since the triangle _DAS_ is similar to the triangle _BAA’_, we may write the proportion

_AB : AD = BA’ : SD._

Also, from the similar triangles _DSC_ and _BCC’_, we have

_CD : CB = SD : B’C._

From these two proportions we have, remembering that _BA’ = BC’_,

[formula]

the minus sign being given to the ratio on account of the fact that _A_ and _C_ are always separated from _B_ and _D_, so that one or three of the segments _AB_, _CD_, _AD_, _CB_ must be negative.

45. Writing the last equation in the form

$$_CB : AB = -CD : AD,_$$

and using the fundamental relation connecting three points on a line,

$$_PR + RQ = PQ,_$$

which holds for all positions of the three points if account be taken of the sign of the segments, the last proportion may be written

$$_(CB - BA) : AB = -(CA - DA) : AD,_$$

or

$$_(AB - AC) : AB = (AC - AD) : AD;_$$

so that _AB_, _AC_, and _AD_ are three quantities in hamonic progression, since the difference between the first and second is to the first as the difference between the second and third is to the third. Also, from this last proportion comes the familiar relation

[formula]

which is convenient for the computation of the distance _AD_ when _AB_ and _AC_ are given numerically.

46. Anharmonic ratio. The corresponding relations between the trigonometric functions of the angles determined by four harmonic lines are not difficult to obtain, but as we shall not need them in building up the theory of projective geometry, we will not discuss them here. Students who have a slight acquaintance with trigonometry may read in a later chapter (§ 161) a development of the theory of a more general relation, called the _anharmonic ratio_, or _cross ratio_, which connects any four points on a line.

PROBLEMS

1. Draw through a given point a line which shall pass through the inaccessible point of intersection of two given lines. The following construction may be made to depend upon Desargues's theorem: Through the given point _P_ draw any two rays cutting the two lines in the points _AB'_ and _A'B_, _A_, _B_, lying on one of the given lines and _A'_, _B'_, on the other. Join _AA'_ and _BB'_, and find their point of intersection

_S_. Through _S_ draw any other ray, cutting the given lines in _CC’_. Join _BC’_ and _B’C_, and obtain their point of intersection _Q_. _PQ_ is the desired line. Justify this construction.

2. To draw through a given point _P_ a line which shall meet two given lines in points _A_ and _B_, equally distant from _P_. Justify the following construction: Join _P_ to the point _S_ of intersection of the two given lines. Construct the fourth harmonic of _PS_ with respect to the two given lines. Draw through _P_ a line parallel to this line. This is the required line.

3. Given a parallelogram in the same plane with a given segment _AC_, to construct linearly the middle point of _AC_.

4. Given four harmonic lines, of which one pair are at right angles to each other, show that the other pair make equal angles with them. This is a theorem of which frequent use will be made.

5. Given the middle point of a line segment, to draw a line parallel to the segment and passing through a given point.

6. A line is drawn cutting the sides of a triangle _ABC_ in the points _A’_, _B’_, _C’_ the point _A’_ lying on the side _BC_, etc. The harmonic conjugate of _A’_ with respect to _B_ and _C_ is then constructed and called _A"_. Similarly, _B"_ and _C"_ are constructed. Show that _A"B"C"_ lie on a straight line. Find other sets of three points on a line in the figure. Find also sets of three lines through a point.

CHAPTER III - COMBINATION OF TWO PROJECTIVELY RELATED FUNDAMENTAL FORMS

[Figure 9]

FIG. 9

47. Superposed fundamental forms. Self-corresponding elements. We have seen (§ 37) that two projective point-rows may be superposed upon the same straight line. This happens, for example, when two pencils which are projective to each other are cut across by a straight line. It is also possible for two projective pencils to have the same center. This happens, for example, when two projective point-rows are projected to the same point. Similarly, two projective axial pencils may have the same axis. We examine now the possibility of two forms related in this way, having an element or elements that correspond to themselves. We have seen, indeed, that if _B_ and _D_ are harmonic conjugates with respect to _A_ and _C_, then the point-row described by _B_ is projective to the point-row described by _D_, and that _A_ and _C_ are self-corresponding points. Consider more generally the case of two pencils perspective to each other with axis of perspectivity _u’_ (Fig. 9). Cut across them by a line _u_. We get thus two projective point-rows superposed on the same line _u_, and a moment’s reflection serves to show that the point _N_ of intersection _u_ and _u’_ corresponds to itself in the two point-rows. Also, the point _M_, where _u_ intersects

the line joining the centers of the two pencils, is seen to correspond to itself. It is thus possible for two projective point-rows, superposed upon the same line, to have two self-corresponding points. Clearly _M_ and _N_ may fall together if the line joining the centers of the pencils happens to pass through the point of intersection of the lines _u_ and _u'_.

[Figure 10]

FIG. 10

48. We may also give an illustration of a case where two superposed projective point-rows have no self-corresponding points at all. Thus we may take two lines revolving about a fixed point _S_ and always making the same angle a with each other (Fig. 10). They will cut out on any line _u_ in the plane two point-rows which are easily seen to be projective. For, given any four rays _SP_ which are harmonic, the four corresponding rays _SP'_ must also be harmonic, since they make the same angles with each other. Four harmonic points _P_ correspond, therefore, to four harmonic points _P'_. It is clear, however, that no point _P_ can coincide with its corresponding point _P'_, for in that case the lines _PS_ and _P'S_ would coincide, which is impossible if the angle between them is to be constant.

49. Fundamental theorem. Postulate of continuity. We have thus shown that two projective point-rows, superposed one on the other, may have two points, one point, or no point at all corresponding to themselves. We proceed to show that

If two projective point-rows, superposed upon the same straight line, have more than two self-corresponding points, they must have an infinite number, and every point corresponds to itself; that is, the two point-rows are not essentially distinct.

If three points, _A_, _B_, and _C_, are self-corresponding, then the harmonic conjugate _D_ of _B_ with respect to _A_ and _C_ must also correspond to itself. For four harmonic points must always correspond to four harmonic points. In the same way the harmonic conjugate of _D_ with respect to _B_ and _C_ must correspond to itself. Combining new points with old in this way, we may obtain as many self-corresponding points as we wish. We show further that every point on the line is the limiting point of a finite or infinite sequence of self-corresponding points. Thus, let a point _P_ lie between _A_ and _B_. Construct now _D_, the fourth harmonic of _C_ with respect to _A_ and _B_. _D_ may coincide with _P_, in which case the sequence is closed; otherwise _P_ lies in the stretch _AD_ or in the stretch _DB_. If it lies in the stretch _DB_, construct the fourth harmonic of _C_ with respect to _D_ and _B_. This point _D'_ may coincide with _P_, in which case, as before, the sequence is closed. If _P_ lies in the stretch _DD'_, we construct the fourth harmonic of _C_ with respect to _DD'_, etc. In each step the region in which _P_ lies is diminished, and the process may be continued until two self-corresponding points are obtained on either side of _P_, and at distances from it arbitrarily small.

We now assume, explicitly, the fundamental postulate that the correspondence is _continuous_, that is, that _the distance between two points in one point-row may be

made arbitrarily small by sufficiently diminishing the distance between the corresponding points in the other._ Suppose now that _P_ is not a self-corresponding point, but corresponds to a point _P'_ at a fixed distance _d_ from _P_. As noted above, we can find self-corresponding points arbitrarily close to _P_, and it appears, then, that we can take a point _D_ as close to _P_ as we wish, and yet the distance between the corresponding points _D'_ and _P'_ approaches _d_ as a limit, and not zero, which contradicts the postulate of continuity.

50. It follows also that two projective pencils which have the same center may have no more than two self-corresponding rays, unless the pencils are identical. For if we cut across them by a line, we obtain two projective point-rows superposed on the same straight line, which may have no more than two self-corresponding points. The same considerations apply to two projective axial pencils which have the same axis.

51. Projective point-rows having a self-corresponding point in common. Consider now two projective point-rows lying on different lines in the same plane. Their common point may or may not be a self-corresponding point. If the two point-rows are perspectively related, then their common point is evidently a self-corresponding point. The converse is also true, and we have the very important theorem:

52. _If in two protective point-rows, the point of intersection corresponds to itself, then the point-rows are in perspective position._

[Figure 11]

FIG. 11

Let the two point-rows be _u_ and _u'_ (Fig. 11). Let _A_ and _A'_, _B_ and _B'_, be corresponding points, and let also the point _M_ of intersection of _u_ and _u'_ correspond to itself. Let _AA'_ and _BB'_ meet in the point _S_. Take _S_ as the center of two pencils, one perspective to _u_ and the other perspective to _u'_. In these two pencils _SA_ coincides with its corresponding ray _SA'_, _SB_ with its corresponding ray _SB'_, and _SM_ with its corresponding ray _SM'_. The two pencils are thus identical, by the preceding theorem, and any ray _SD_ must coincide with its corresponding ray _SD'_. Corresponding points of _u_ and _u'_, therefore, all lie on lines through the point _S_.

53. An entirely similar discussion shows that

If in two projective pencils the line joining their centers is a self-corresponding ray, then the two pencils are perspectively related.

54. A similar theorem may be stated for two axial pencils of which the axes intersect. Very frequent use will be made of these fundamental theorems.

55. Point-row of the second order. The question naturally arises, What is the locus of points of intersection of corresponding rays of two projective pencils which are not in perspective position? This locus, which will be discussed in detail in subsequent chapters, is easily seen to have at most two points in common with any line in the plane, and on account of this fundamental property will be called a

point-row of the second order. For any line _u_ in the plane of the two pencils will be cut by them in two projective point-rows which have at most two self-corresponding points. Such a self-corresponding point is clearly a point of intersection of corresponding rays of the two pencils.

56. This locus degenerates in the case of two perspective pencils to a pair of straight lines, one of which is the axis of perspectivity and the other the common ray, any point of which may be considered as the point of intersection of corresponding rays of the two pencils.

57. Pencils of rays of the second order. Similar investigations may be made concerning the system of lines joining corresponding points of two projective point-rows. If we project the point-rows to any point in the plane, we obtain two projective pencils having the same center. At most two pairs of self-corresponding rays may present themselves. Such a ray is clearly a line joining two corresponding points in the two point-rows. The result may be stated as follows: _The system of rays joining corresponding points in two protective point-rows has at most two rays in common with any pencil in the plane._ For that reason the system of rays is called _a pencil of rays of the second order._

58. In the case of two perspective point-rows this system of rays degenerates into two pencils of rays of the first order, one of which has its center at the center of perspectivity of the two point-rows, and the other at the intersection of the two point-rows, any ray through which may be considered as joining two corresponding points of the two point-rows.

59. Cone of the second order. The corresponding theorems in space may easily be obtained by joining the points and lines considered in the plane theorems to a point _S_ in space. Two projective pencils give rise to two projective axial pencils with axes intersecting. Corresponding planes meet in lines which all pass through _S_ and through the points on a point-row of the second order generated by the two pencils of rays. They are thus generating lines of a _cone of the second order_, or _quadric cone_, so called because every plane in space not passing through _S_ cuts it in a point-row of the second order, and every line also cuts it in at most two points. If, again, we project two point-rows to a point _S_ in space, we obtain two pencils of rays with a common center but lying in different planes. Corresponding lines of these pencils determine planes which are the projections to _S_ of the lines which join the corresponding points of the two point-rows. At most two such planes may pass through any ray through _S_. It is called _a pencil of planes of the second order_.

PROBLEMS

*1. * A man _A_ moves along a straight road _u_, and another man _B_ moves along the same road and walks so as always to keep sight of _A_ in a small mirror _M_ at the side of the road. How many times will they come together, _A_ moving always in the same direction along the road?

2. How many times would the two men in the first problem see each other in two mirrors _M_ and _N_ as they walk along the road as before? (The planes of the two mirrors are not necessarily parallel to _u_.)

3. As A moves along _u_, trace the path of B so that the two men may always see each other in the two mirrors.

4. Two boys walk along two paths _u_ and _u'_ each holding a string which they keep stretched tightly between them. They both move at constant but different rates of speed, letting out the string or drawing it in as they walk. How many times will the line of the string pass over any given point in the plane of the paths?

5. Trace the lines of the string when the two boys move at the same rate of speed in the two paths but do not start at the same time from the point where the two paths intersect.

6. A ship is sailing on a straight course and keeps a gun trained on a point on the shore. Show that a line at right angles to the direction of the gun at its muzzle will pass through any point in the plane twice or not at all. (Consider the point-row at infinity cut out by a line through the point on the shore at right angles to the direction of the gun.)

7. Two lines _u_ and _u'_ revolve about two points _U_ and _U'_ respectively in the same plane. They go in the same direction and at the same rate of speed, but one has an angle a the start of the other. Show that they generate a point-row of the second order.

8. Discuss the question given in the last problem when the two lines revolve in opposite directions. Can you recognize the locus?

CHAPTER IV - POINT-ROWS OF THE SECOND ORDER

60. Point-row of the second order defined. We have seen that two fundamental forms in one-to-one correspondence may sometimes generate a form of higher order. Thus, two point-rows (§ 55) generate a system of rays of the second order, and two pencils of rays (§ 57), a system of points of the second order. As a system of points is more familiar to most students of geometry than a system of lines, we study first the point-row of the second order.

61. Tangent line. We have shown in the last chapter (§ 55) that the locus of intersection of corresponding rays of two projective pencils is a point-row of the second order; that is, it has at most two points in common with any line in the plane. It is clear, first of all, that the centers of the pencils are points of the locus; for to the line _SS'_, considered as a ray of _S_, must correspond some ray of _S'_ which meets it in _S'_. _S'_, and by the same argument _S_, is then a point where corresponding rays meet. Any ray through _S_ will meet it in one point besides _S_, namely, the point _P_ where it meets its corresponding ray. Now, by choosing the ray through _S_ sufficiently close to the ray _SS'_, the point _P_ may be made to approach arbitrarily close to _S'_, and the ray _S'P_ may be made to differ in position from the tangent line at _S'_ by as little as we please. We have, then, the important theorem

_The ray at __S’__ which corresponds to the common ray __SS’__ is tangent to the locus at __S’__._

In the same manner the tangent at _S_ may be constructed.

62. Determination of the locus. We now show that _it is possible to assign arbitrarily the position of three points, __A__, __B__, and __C__, on the locus (besides the points __S__ and __S’__); but, these three points being chosen, the locus is completely determined._

63. This statement is equivalent to the following:

Given three pairs of corresponding rays in two projective pencils, it is possible to find a ray of one which corresponds to any ray of the other.

64. We proceed, then, to the solution of the fundamental

PROBLEM: _Given three pairs of rays, __aa’__, __bb’__, and __cc’__, of two protective pencils, __S__ and __S’__, to find the ray __d’__ of __S’__ which corresponds to any ray __d__ of __S__._

[Figure 12]

FIG. 12

Call _A_ the intersection of _aa’_, _B_ the intersection of _bb’_, and _C_ the intersection of _cc’_ (Fig. 12). Join _AB_ by the line _u_, and _AC_ by the line _u’_. Consider _u_ as a point-row perspective to _S_, and _u’_ as a point-row perspective to _S’_. _u_ and _u’_ are projectively related to each other, since _S_ and _S’_ are, by hypothesis, so related. But their point of intersection _A_ is a self-corresponding point, since _a_ and _a’_ were supposed to be corresponding rays. It follows (§ 52) that _u_ and _u’_ are in perspective position, and that lines through corresponding points all pass through a point _M_, the center of perspectivity, the position of which will be determined by any two such lines. But the intersection of _a_ with _u_ and the intersection of _c’_ with _u’_ are corresponding points on _u_ and _u’_, and the line joining them is clearly _c_ itself. Similarly, _b’_ joins two corresponding points on _u_ and _u’_, and so the center _M_ of perspectivity of _u_ and _u’_ is the intersection of _c_ and _b’_. To find _d’_ in _S’_ corresponding to a given line _d_ of _S_ we note the point _L_ where _d_ meets _u_. Join _L_ to _M_ and get the point _N_ where this line meets _u’_. _L_ and _N_ are corresponding points on _u_ and _u’_, and _d’_ must therefore pass through _N_. The intersection _P_ of _d_ and _d’_ is thus another point on the locus. In the same manner any number of other points may be obtained.

65. The lines _u_ and _u’_ might have been drawn in any direction through _A_ (avoiding, of course, the line _a_ for _u_ and the line _a’_ for _u’_), and the center of perspectivity _M_ would be easily obtainable; but the above construction furnishes a simple and instructive figure. An equally simple one is obtained by taking _a’_ for _u_ and _a_ for _u’_.

66. Lines joining four points of the locus to a fifth. Suppose that the points _S_, _S’_, _B_, _C_, and _D_ are fixed, and that four points, _A_, _A__1_, _A__2_, and _A__3_,

are taken on the locus at the intersection with it of any four harmonic rays through _B_. These four harmonic rays give four harmonic points, _L_, _L__1_ etc., on the fixed ray _SD_. These, in turn, project through the fixed point _M_ into four harmonic points, _N_, _N__1_ etc., on the fixed line _DS'_. These last four harmonic points give four harmonic rays _CA_, _CA__1_, _CA__2_, _CA__3_. Therefore the four points _A_ which project to _B_ in four harmonic rays also project to _C_ in four harmonic rays. But _C_ may be any point on the locus, and so we have the very important theorem,

Four points which are on the locus, and which project to a fifth point of the locus in four harmonic rays, project to any point of the locus in four harmonic rays.

67. The theorem may also be stated thus:

The locus of points from which, four given points are seen along four harmonic rays is a point-row of the second order through them.

68. A further theorem of prime importance also follows:

Any two points on the locus may be taken as the centers of two projective pencils which will generate the locus.

69. Pascal's theorem. The points _A_, _B_, _C_, _D_, _S_, and _S'_ may thus be considered as chosen arbitrarily on the locus, and the following remarkable theorem follows at once.

Given six points, 1, 2, 3, 4, 5, 6, on the point-row of the second order, if we call

L the intersection of 12 with 45,

M the intersection of 23 with 56,

N the intersection of 34 with 61,

_then __L__, __M__, and __N__ are on a straight line._

[Figure 13]

FIG. 13

70. To get the notation to correspond to the figure, we may take (Fig. 13) _A = 1_, _B = 2_, _S' = 3_, _D = 4_, _S = 5_, and _C = 6_. If we make _A = 1_, _C=2_, _S=3_, _D = 4_, _S'=5_, and. _B = 6_, the points _L_ and _N_ are interchanged, but the line is left unchanged. It is clear that one point may be named arbitrarily and the other five named in _5! = 120_ different ways, but since, as we have seen, two different assignments of names give the same line, it follows that there cannot be more than 60 different lines _LMN_ obtained in this way from a given set of six points. As a matter of fact, the number obtained in this way is in general _60_. The above theorem, which is of cardinal importance in the theory of the point-row of the second order, is due to Pascal and was discovered by him at the age of sixteen. It is, no doubt, the most important contribution to the theory of these loci since the days of Apollonius. If the six points be called the vertices of a hexagon inscribed in the curve, then the sides 12 and 45 may be appropriately called a pair of opposite sides. Pascal's theorem, then,

may be stated as follows:

The three pairs of opposite sides of a hexagon inscribed in a point-row of the second order meet in three points on a line.

71. Harmonic points on a point-row of the second order. Before proceeding to develop the consequences of this theorem, we note another result of the utmost importance for the higher developments of pure geometry, which follows from the fact that if four points on the locus project to a fifth in four harmonic rays, they will project to any point of the locus in four harmonic rays. It is natural to speak of four such points as four harmonic points on the locus, and to use this notion to define projective correspondence between point-rows of the second order, or between a point-row of the second order and any fundamental form of the first order. Thus, in particular, the point-row of the second order, σ, is said to be _perspectively related_ to the pencil _S_ when every ray on _S_ goes through the point on σ which corresponds to it.

72. Determination of the locus. It is now clear that five points, arbitrarily chosen in the plane, are sufficient to determine a point-row of the second order through them. Two of the points may be taken as centers of two projective pencils, and the three others will determine three pairs of corresponding rays of the pencils, and therefore all pairs. If four points of the locus are given, together with the tangent at one of them, the locus is likewise completely determined. For if the point at which the tangent is given be taken as the center _S_ of one pencil, and any other of the points for _S'_, then, besides the two pairs of corresponding rays determined by the remaining two points, we have one more pair, consisting of the tangent at _S_ and the ray _SS'_. Similarly, the curve is determined by three points and the tangents at two of them.

73. Circles and conics as point-rows of the second order. It is not difficult to see that a circle is a point-row of the second order. Indeed, take any point _S_ on the circle and draw four harmonic rays through it. They will cut the circle in four points, which will project to any other point of the curve in four harmonic rays; for, by the theorem concerning the angles inscribed in a circle, the angles involved in the second set of four lines are the same as those in the first set. If, moreover, we project the figure to any point in space, we shall get a cone, standing on a circular base, generated by two projective axial pencils which are the projections of the pencils at _S_ and _S'_. Cut across, now, by any plane, and we get a conic section which is thus exhibited as the locus of intersection of two projective pencils. It thus appears that a conic section is a point-row of the second order. It will later appear that a point-row of the second order is a conic section. In the future, therefore, we shall refer to a point-row of the second order as a conic.

[Figure 14]

FIG. 14

74. Conic through five points. Pascal's theorem furnishes an elegant solution of the problem of drawing a conic through five given points. To construct a sixth point on the conic, draw through the point numbered 1 an arbitrary line (Fig. 14), and let the desired point 6 be the second point of intersection of this line with the conic. The point _L = 12-45_ is obtainable at once; also the point _N = 34-61_. But _L_ and _N_ determine Pascal's line, and the intersection of 23 with 56 must be on this line. Intersect, then, the line _LN_ with 23 and obtain the point _M_. Join _M_ to 5 and intersect with 61 for the desired point 6.

[Figure 15]

FIG. 15

75. Tangent to a conic. If two points of Pascal's hexagon approach coincidence, then the line joining them approaches as a limiting position the tangent line at that point. Pascal's theorem thus affords a ready method of drawing the tangent line to a conic at a given point. If the conic is determined by the points 1, 2, 3, 4, 5 (Fig. 15), and it is desired to draw the tangent at the point 1, we may call that point 1, 6. The points _L_ and _M_ are obtained as usual, and the intersection of 34 with _LM_ gives _N_. Join _N_ to the point 1 for the desired tangent at that point.

76. Inscribed quadrangle. Two pairs of vertices may coalesce, giving an inscribed quadrangle. Pascal's theorem gives for this case the very important theorem

Two pairs of opposite sides of any quadrangle inscribed in a conic meet on a straight line, upon which line also intersect the two pairs of tangents at the opposite vertices.

[Figure 16]

FIG. 16

[Figure 17]

FIG. 17

For let the vertices be _A_, _B_, _C_, and _D_, and call the vertex _A_ the point 1, 6; _B_, the point 2; _C_, the point 3, 4; and _D_, the point 5 (Fig. 16). Pascal's theorem then indicates that _L = AB-CD_, _M = AD-BC_, and _N_, which is the intersection of the tangents at _A_ and _C_, are all on a straight line _u_. But if we were to call _A_ the point 2, _B_ the point 6, 1, _C_ the point 5, and _D_ the point 4, 3, then the intersection _P_ of the tangents at _B_ and _D_ are also on this same line _u_. Thus _L_, _M_, _N_, and _P_ are four points on a straight line. The consequences of this theorem are so numerous and important that we shall devote a separate chapter to them.

77. Inscribed triangle. Finally, three of the vertices of the hexagon may coalesce, giving a triangle inscribed in a conic. Pascal's theorem then reads as follows (Fig. 17)

for this case:

The three tangents at the vertices of a triangle inscribed in a conic meet the opposite sides in three points on a straight line.

[Figure 18]

FIG. 18

78. Degenerate conic. If we apply Pascal's theorem to a degenerate conic made up of a pair of straight lines, we get the following theorem (Fig. 18):

_If three points, __A__, __B__, __C__, are chosen on one line, and three points, __A'__, __B'__, __C'__, are chosen on another, then the three points __L = AB'-A'B__, __M = BC'-B'C__, __N = CA'-C'A__ are all on a straight line._

PROBLEMS

1. In Fig. 12, select different lines _u_ and trace the locus of the center of perspectivity _M_ of the lines _u_ and _u'_.

2. Given four points, _A_, _B_, _C_, _D_, in the plane, construct a fifth point _P_ such that the lines _PA_, _PB_, _PC_, _PD_ shall be four harmonic lines.

Suggestion. Draw a line _a_ through the point _A_ such that the four lines _a_, _AB_, _AC_, _AD_ are harmonic. Construct now a conic through _A_, _B_, _C_, and _D_ having _a_ for a tangent at _A_.

3. Where are all the points _P_, as determined in the preceding question, to be found?

4. Select any five points in the plane and draw the tangent to the conic through them at each of the five points.

5. Given four points on the conic, and the tangent at one of them, to construct the conic. ("To construct the conic" means here to construct as many other points as may be desired.)

6. Given three points on the conic, and the tangent at two of them, to construct the conic.

7. Given five points, two of which are at infinity in different directions, to construct the conic. (In this, and in the following examples, the student is supposed to be able to draw a line parallel to a given line.)

8. Given four points on a conic (two of which are at infinity and two in the finite part of the plane), together with the tangent at one of the finite points, to construct the conic.

9. The tangents to a curve at its infinitely distant points are called its _asymptotes_ if they pass through a finite part of the plane. Given the asymptotes and a finite point of a conic, to construct the conic.

10. Given an asymptote and three finite points on the conic, to determine the conic.

11. Given four points, one of which is at infinity, and given also that the line at infinity is a tangent line, to construct the conic.

CHAPTER V - PENCILS OF RAYS OF THE SECOND ORDER

79. Pencil of rays of the second order defined. If the corresponding points of two projective point-rows be joined by straight lines, a system of lines is obtained which is called a pencil of rays of the second order. This name arises from the fact, easily shown (§ 57), that at most two lines of the system may pass through any arbitrary point in the plane. For if through any point there should pass three lines of the system, then this point might be taken as the center of two projective pencils, one projecting one point-row and the other projecting the other. Since, now, these pencils have three rays of one coincident with the corresponding rays of the other, the two are identical and the two point-rows are in perspective position, which was not supposed.

[Figure 19]

FIG. 19

80. Tangents to a circle. To get a clear notion of this system of lines, we may first show that the tangents to a circle form a system of this kind. For take any two tangents, _u_ and _u’_, to a circle, and let _A_ and _B_ be the points of contact (Fig. 19). Let now _t_ be any third tangent with point of contact at _C_ and meeting _u_ and _u’_ in _P_ and _P’_ respectively. Join _A_, _B_, _P_, _P’_, and _C_ to _O_, the center of the circle. Tangents from any point to a circle are equal, and therefore the triangles _POA_ and _POC_ are equal, as also are the triangles _P’OB_ and _P’OC_. Therefore the angle _POP’_ is constant, being equal to half the constant angle _AOC + COB_. This being true, if we take any four harmonic points, _P__1_, _P__2_, _P__3_, _P__4_, on the line _u_, they will project to _O_ in four harmonic lines, and the tangents to the circle from these four points will meet _u’_ in four harmonic points, _P’__1_, _P’__2_, _P’__3_, _P’__4_, because the lines from these points to _O_ inclose the same angles as the lines from the points _P__1_, _P__2_, _P__3_, _P__4_ on _u_. The point-row on _u_ is therefore projective to the point-row on _u’_. Thus the tangents to a circle are seen to join corresponding points on two projective point-rows, and so, according to the definition, form a pencil of rays of the second order.

81. Tangents to a conic. If now this figure be projected to a point outside the plane of the circle, and any section of the resulting cone be made by a plane, we can easily see that the system of rays tangent to any conic section is a pencil of rays of the second order. The converse is also true, as we shall see later, and a pencil of rays of the second order is also a set of lines tangent to a conic section.

82. The point-rows _u_ and _u’_ are, themselves, lines of the system, for to the common point of the two point-rows, considered as a point of _u_, must correspond some point of _u’_, and the line joining these two corresponding points is clearly _u’_

itself. Similarly for the line _u_.

83. Determination of the pencil. We now show that _it is possible to assign arbitrarily three lines, __a__, __b__, and __c__, of __ the system (besides the lines __u__ and __u'__); but if these three lines are chosen, the system is completely determined._

This statement is equivalent to the following:

Given three pairs of corresponding points in two projective point-rows, it is possible to find a point in one which corresponds to any point of the other.

We proceed, then, to the solution of the fundamental

PROBLEM. _Given three pairs of points, __AA'__, __BB'__, and __CC'__, of two projective point-rows __u__ and __u'__, to find the point __D'__ of __u'__ which corresponds to any given point __D__ of __u__._

[Figure 20]

FIG. 20

On the line _a_, joining _A_ and _A'_, take two points, _S_ and _S'_, as centers of pencils perspective to _u_ and _u'_ respectively (Fig. 20). The figure will be much simplified if we take _S_ on _BB'_ and _S'_ on _CC'_. _SA_ and _S'A'_ are corresponding rays of _S_ and _S'_, and the two pencils are therefore in perspective position. It is not difficult to see that the axis of perspectivity _m_ is the line joining _B'_ and _C_. Given any point _D_ on _u_, to find the corresponding point _D'_ on _u'_ we proceed as follows: Join _D_ to _S_ and note where the joining line meets _m_. Join this point to _S'_. This last line meets _u'_ in the desired point _D'_.

We have now in this figure six lines of the system, _a_, _b_, _c_, _d_, _u_, and _u'_. Fix now the position of _u_, _u'_, _b_, _c_, and _d_, and take four lines of the system, _a__1_, _a__2_, _a__3_, _a__4_, which meet _b_ in four harmonic points. These points project to _D_, giving four harmonic points on _m_. These again project to _D'_, giving four harmonic points on _c_. It is thus clear that the rays _a__1_, _a__2_, _a__3_, _a__4_ cut out two projective point-rows on any two lines of the system. Thus _u_ and _u'_ are not special rays, and any two rays of the system will serve as the point-rows to generate the system of lines.

84. Brianchon's theorem. From the figure also appears a fundamental theorem due to Brianchon:

_If __1__, __2__, __3__, __4__, __5__, __6__ are any six rays of a pencil of the second order, then the lines __l = (12, 45)__, __m = (23, 56)__, __n = (34, 61)__ all pass through a point._

[Figure 21]

FIG. 21

85. To make the notation fit the figure (Fig. 21), make _a=1_, _b = 2_, _u’ = 3_, _d = 4_, _u = 5_, _c = 6_; or, interchanging two of the lines, _a = 1_, _c = 2_, _u = 3_, _d = 4_, _u’ = 5_, _b = 6_. Thus, by different namings of the lines, it appears that not more than 60 different _Brianchon points_ are possible. If we call 12 and 45 opposite vertices of a circumscribed hexagon, then Brianchon’s theorem may be stated as follows:

The three lines joining the three pairs of opposite vertices of a hexagon circumscribed about a conic meet in a point.

86. Construction of the pencil by Brianchon’s theorem. Brianchon’s theorem furnishes a ready method of determining a sixth line of the pencil of rays of the second order when five are given. Thus, select a point in line 1 and suppose that line 6 is to pass through it. Then _l = (12, 45)_, _n = (34, 61)_, and the line _m = (23, 56)_ must pass through _(l, n)_. Then _(23, ln)_ meets 5 in a point of the required sixth line.

[Figure 22]

FIG. 22

87. Point of contact of a tangent to a conic. If the line 2 approach as a limiting position the line 1, then the intersection _(1, 2)_ approaches as a limiting position the point of contact of 1 with the conic. This suggests an easy way to construct the point of contact of any tangent with the conic. Thus (Fig. 22), given the lines 1, 2, 3, 4, 5 to construct the point of contact of _1=6_. Draw _l = (12,45)_, _m =(23,56)_; then _(34, lm)_ meets 1 in the required point of contact _T_.

[Figure 23]

FIG. 23

88. Circumscribed quadrilateral. If two pairs of lines in Brianchon’s hexagon coalesce, we have a theorem concerning a quadrilateral circumscribed about a conic. It is easily found to be (Fig. 23)

The four lines joining the two opposite pairs of vertices and the two opposite points of contact of a quadrilateral circumscribed about a conic all meet in a point. The consequences of this theorem will be deduced later.

[Figure 24]

FIG. 24

89. Circumscribed triangle. The hexagon may further degenerate into a triangle, giving the theorem (Fig. 24) _The lines joining the vertices to the points of contact of the opposite sides of a triangle circumscribed about a conic all meet in a point._

90. Brianchon’s theorem may also be used to solve the following problems:

Given four tangents and the point of contact on any one of them, to construct other tangents to a conic. Given three tangents and the points of contact of any two of them, to construct other tangents to a conic.

91. Harmonic tangents. We have seen that a variable tangent cuts out on any two fixed tangents projective point-rows. It follows that if four tangents cut a fifth in four harmonic points, they must cut every tangent in four harmonic points. It is possible, therefore, to make the following definition:

Four tangents to a conic are said to be harmonic when they meet every other tangent in four harmonic points.

92. Projectivity and perspectivity. This definition suggests the possibility of defining a projective correspondence between the elements of a pencil of rays of the second order and the elements of any form heretofore discussed. In particular, the points on a tangent are said to be _perspectively related_ to the tangents of a conic when each point lies on the tangent which corresponds to it. These notions are of importance in the higher developments of the subject.

[Figure 25]

FIG. 25

93. Brianchon’s theorem may also be applied to a degenerate conic made up of two points and the lines through them. Thus(Fig. 25),

_If __a__, __b__, __c__ are three lines through a point __S__, and __a’__, __b’__, __c’__ are three lines through another point __S’__, then the lines __l = (ab’, a’b)__, __m = (bc’, b’c)__, and __n = (ca’, c’a)__ all meet in a point._

94. Law of duality. The observant student will not have failed to note the remarkable similarity between the theorems of this chapter and those of the preceding. He will have noted that points have replaced lines and lines have replaced points; that points on a curve have been replaced by tangents to a curve; that pencils have been replaced by point-rows, and that a conic considered as made up of a succession of points has been replaced by a conic considered as generated by a moving tangent line. The theory upon which this wonderful _law of duality_ is based will be developed in the next chapter.

PROBLEMS

1. Given four lines in the plane, to construct another which shall meet them in four harmonic points.

2. Where are all such lines found?

3. Given any five lines in the plane, construct on each the point of contact with the conic tangent to them all.

4. Given four lines and the point of contact on one, to construct the conic. ("To construct the conic" means here to draw as many other tangents as may be desired.)

5. Given three lines and the point of contact on two of them, to construct the conic.

6. Given four lines and the line at infinity, to construct the conic.

7. Given three lines and the line at infinity, together with the point of contact at infinity, to construct the conic.

8. Given three lines, two of which are asymptotes, to construct the conic.

9. Given five tangents to a conic, to draw a tangent which shall be parallel to any one of them.

10. The lines _a_, _b_, _c_ are drawn parallel to each other. The lines _a'_, _b'_, _c'_ are also drawn parallel to each other. Show why the lines (_ab'_, _a'b_), (_bc'_, _b'c_), (_ca'_, _c'a_) meet in a point. (In problems 6 to 10 inclusive, parallel lines are to be drawn.)

CHAPTER VI - POLES AND POLARS

95. Inscribed and circumscribed quadrilaterals. The following theorems have been noted as special cases of Pascal's and Brianchon's theorems:

If a quadrilateral be inscribed in a conic, two pairs of opposite sides and the tangents at opposite vertices intersect in four points, all of which lie on a straight line.

If a quadrilateral be circumscribed about a conic, the lines joining two pairs of opposite vertices and the lines joining two opposite points of contact are four lines which meet in a point.

[Figure 26]

FIG. 26

96. Definition of the polar line of a point. Consider the quadrilateral _K_, _L_, _M_, _N_ inscribed in the conic (Fig. 26). It determines the four harmonic points _A_, _B_, _C_, _D_ which project from _N_ in to the four harmonic points _M_, _B_, _K_, _O_. Now the tangents at _K_ and _M_ meet in _P_, a point on the line _AB_. The line _AB_ is thus determined entirely by the point _O_. For if we draw any line through it, meeting the conic in _K_ and _M_, and construct the harmonic conjugate _B_ of _O_ with respect to _K_ and _M_, and also the two tangents at _K_ and _M_ which meet in the point _P_, then _BP_ is the line in question. It thus appears that the line _LON_ may be any line whatever through _O_; and since _D_, _L_, _O_, _N_ are four harmonic points, we may describe the line _AB_ as the locus of points which are harmonic conjugates of _O_ with respect to the two points where any line through _O_ meets the curve.

97. Furthermore, since the tangents at _L_ and _N_ meet on this same line, it appears as the locus of intersections of pairs of tangents drawn at the extremities of chords through _O_.

98. This important line, which is completely determined by the point _O_, is called the _polar_ of _O_ with respect to the conic; and the point _O_ is called the _pole_ of the line with respect to the conic.

99. If a point _B_ is on the polar of _O_, then it is harmonically conjugate to _O_ with respect to the two intersections _K_ and _M_ of the line _BC_ with the conic. But for the same reason _O_ is on the polar of _B_. We have, then, the fundamental theorem

If one point lies on the polar of a second, then the second lies on the polar of the first.

100. Conjugate points and lines. Such a pair of points are said to be _conjugate_ with respect to the conic. Similarly, lines are said to be _conjugate_ to each other with respect to the conic if one, and consequently each, passes through the pole of the other.

[Figure 27]

FIG. 27

101. Construction of the polar line of a given point. Given a point _P_, if it is within the conic (that is, if no tangents may be drawn from _P_ to the conic), we may construct its polar line by drawing through it any two chords and joining the two points of intersection of the two pairs of tangents at their extremities. If the point _P_ is outside the conic, we may draw the two tangents and construct the chord of contact (Fig. 27).

102. Self-polar triangle. In Fig. 26 it is not difficult to see that _AOC_ is a _self-polar_ triangle, that is, each vertex is the pole of the opposite side. For _B_, _M_, _O_, _K_ are four harmonic points, and they project to _C_ in four harmonic rays. The line _CO_, therefore, meets the line _AMN_ in a point on the polar of _A_, being separated from _A_ harmonically by the points _M_ and _N_. Similarly, the line _CO_ meets _KL_ in a point on the polar of _A_, and therefore _CO_ is the polar of _A_. Similarly, _OA_ is the polar of _C_, and therefore _O_ is the pole of _AC_.

103. Pole and polar projectively related. Another very important theorem comes directly from Fig. 26.

_As a point __A__ moves along a straight line its polar with respect to a conic revolves about a fixed point and describes a pencil projective to the point-row described by __A__._

For, fix the points _L_ and _N_ and let the point _A_ move along the line _AQ_; then the point-row _A_ is projective to the pencil _LK_, and since _K_ moves along the conic, the pencil _LK_ is projective to the pencil _NK_, which in turn is projective to the point-row _C_, which, finally, is projective to the pencil _OC_, which is the polar of _A_.

104. Duality. We have, then, in the pole and polar relation a device for setting up a one-to-one correspondence between the points and lines of the plane—a

correspondence which may be called projective, because to four harmonic points or lines correspond always four harmonic lines or points. To every figure made up of points and lines will correspond a figure made up of lines and points. To a point-row of the second order, which is a conic considered as a point-locus, corresponds a pencil of rays of the second order, which is a conic considered as a line-locus. The name ’duality’ is used to describe this sort of correspondence. It is important to note that the dual relation is subject to the same exceptions as the one-to-one correspondence is, and must not be appealed to in cases where the one-to-one correspondence breaks down. We have seen that there is in Euclidean geometry one and only one ray in a pencil which has no point in a point-row perspective to it for a corresponding point; namely, the line parallel to the line of the point-row. Any theorem, therefore, that involves explicitly the point at infinity is not to be translated into a theorem concerning lines. Further, in the pencil the angle between two lines has nothing to correspond to it in a point-row perspective to the pencil. Any theorem, therefore, that mentions angles is not translatable into another theorem by means of the law of duality. Now we have seen that the notion of the infinitely distant point on a line involves the notion of dividing a segment into any number of equal parts—in other words, of _measuring_. If, therefore, we call any theorem that has to do with the line at infinity or with the measurement of angles a _metrical_ theorem, and any other kind a _projective_ theorem, we may put the case as follows:

Any projective theorem involves another theorem, dual to it, obtainable by interchanging everywhere the words ’point’ and ’line.’

105. Self-dual theorems. The theorems of this chapter will be found, upon examination, to be _self-dual_; that is, no new theorem results from applying the process indicated in the preceding paragraph. It is therefore useless to look for new results from the theorem on the circumscribed quadrilateral derived from Brianchon’s, which is itself clearly the dual of Pascal’s theorem, and in fact was first discovered by dualization of Pascal’s.

106. It should not be inferred from the above discussion that one-to-one correspondences may not be devised that will control certain of the so-called metrical relations. A very important one may be easily found that leaves angles unaltered. The relation called _similarity_ leaves ratios between corresponding segments unaltered. The above statements apply only to the particular one-to-one correspondence considered.

PROBLEMS

1. Given a quadrilateral, construct the quadrangle polar to it with respect to a given conic.

2. A point moves along a straight line. Show that its polar lines with respect to two given conics generate a point-row of the second order.

3. Given five points, draw the polar of a point with respect to the conic passing through them, without drawing the conic itself.

4. Given five lines, draw the polar of a point with respect to the conic tangent to them, without drawing the conic itself.

5. Dualize problems 3 and 4.

6. Given four points on the conic, and the tangent at one of them, draw the polar of a given point without drawing the conic. Dualize.

7. A point moves on a conic. Show that its polar line with respect to another conic describes a pencil of rays of the second order.

Suggestion. Replace the given conic by a pair of protective pencils.

8. Show that the poles of the tangents of one conic with respect to another lie on a conic.

9. The polar of a point _A_ with respect to one conic is _a_, and the pole of _a_ with respect to another conic is _A'_. Show that as _A_ travels along a line, _A'_ also travels along another line. In general, if _A_ describes a curve of degree _n_, show that _A'_ describes another curve of the same degree _n_. (The degree of a curve is the greatest number of points that it may have in common with any line in the plane.)

CHAPTER VII - METRICAL PROPERTIES OF THE CONIC SECTIONS

107. Diameters. Center. After what has been said in the last chapter one would naturally expect to get at the metrical properties of the conic sections by the introduction of the infinite elements in the plane. Entering into the theory of poles and polars with these elements, we have the following definitions:

The polar line of an infinitely distant point is called a _diameter_, and the pole of the infinitely distant line is called the _center_, of the conic.

108. From the harmonic properties of poles and polars,

The center bisects all chords through it (§ 39).

Every diameter passes through the center.

All chords through the same point at infinity (that is, each of a set of parallel chords) are bisected by the diameter which is the polar of that infinitely distant point.

109. Conjugate diameters. We have already defined conjugate lines as lines which pass each through the pole of the other (§ 100).

Any diameter bisects all chords parallel to its conjugate.

The tangents at the extremities of any diameter are parallel, and parallel to the conjugate diameter.

Diameters parallel to the sides of a circumscribed parallelogram are conjugate.

All these theorems are easy exercises for the student.

110. Classification of conics. Conics are classified according to their relation to the infinitely distant line. If a conic has two points in common with the line at infinity, it is called a _hyperbola_; if it has no point in common with the infinitely distant line, it is called an _ellipse_; if it is tangent to the line at infinity, it is called a _parabola_.

111. _In a hyperbola the center is outside the curve_ (§ 101), since the two tangents to the curve at the points where it meets the line at infinity determine by their intersection the center. As previously noted, these two tangents are called the _asymptotes_ of the curve. The ellipse and the parabola have no asymptotes.

112. _The center of the parabola is at infinity, and therefore all its diameters are parallel,_ for the pole of a tangent line is the point of contact.

The locus of the middle points of a series of parallel chords in a parabola is a diameter, and the direction of the line of centers is the same for all series of parallel chords.

The center of an ellipse is within the curve.

[Figure 28]

FIG. 28

113. Theorems concerning asymptotes. We derived as a consequence of the theorem of Brianchon (§ 89) the proposition that if a triangle be circumscribed about a conic, the lines joining the vertices to the points of contact of the opposite sides all meet in a point. Take, now, for two of the tangents the asymptotes of a hyperbola, and let any third tangent cut them in _A_ and _B_ (Fig. 28). If, then, _O_ is the intersection of the asymptotes,—and therefore the center of the curve,— then the triangle _OAB_ is circumscribed about the curve. By the theorem just quoted, the line through _A_ parallel to _OB_, the line through _B_ parallel to _OA_, and the line _OP_ through the point of contact of the tangent _AB_ all meet in a point _C_. But _OACB_ is a parallelogram, and _PA = PB_. Therefore

The asymptotes cut off on each tangent a segment which is bisected by the point of contact.

114. If we draw a line _OQ_ parallel to _AB_, then _OP_ and _OQ_ are conjugate diameters, since _OQ_ is parallel to the tangent at the point where _OP_ meets the curve. Then, since _A_, _P_, _B_, and the point at infinity on _AB_ are four harmonic points, we have the theorem

Conjugate diameters of the hyperbola are harmonic conjugates with respect to the asymptotes.

115. The chord _A"B"_, parallel to the diameter _OQ_, is bisected at _P'_ by the conjugate diameter _OP_. If the chord _A"B"_ meet the asymptotes in _A'_, _B'_, then _A'_, _P'_, _B'_, and the point at infinity are four harmonic points, and therefore _P'_ is the middle point of _A'B'_. Therefore _A'A" = B'B"_ and we have the theorem

The segments cut off on any chord between the hyperbola and its asymptotes are equal.

116. This theorem furnishes a ready means of constructing the hyperbola by points when a point on the curve and the two asymptotes are given.

[Figure 29]

FIG. 29

117. For the circumscribed quadrilateral, Brianchon's theorem gave (§ 88) _The lines joining opposite vertices and the lines joining opposite points of contact are four lines meeting in a point._ Take now for two of the tangents the asymptotes, and let _AB_ and _CD_ be any other two (Fig. 29). If _B_ and _D_ are opposite vertices, and also _A_ and _C_, then _AC_ and _BD_ are parallel, and parallel to _PQ_, the line joining the points of contact of _AB_ and _CD_, for these are three of the four lines of the theorem just quoted. The fourth is the line at infinity which joins the point of contact of the asymptotes. It is thus seen that the triangles _ABC_ and _ADC_ are equivalent, and therefore the triangles _AOB_ and _COD_ are also. The tangent AB may be fixed, and the tangent _CD_ chosen arbitrarily; therefore

The triangle formed by any tangent to the hyperbola and the two asymptotes is of constant area.

118. Equation of hyperbola referred to the asymptotes. Draw through the point of contact _P_ of the tangent _AB_ two lines, one parallel to one asymptote and the other parallel to the other. One of these lines meets _OB_ at a distance _y_ from _O_, and the other meets _OA_ at a distance _x_ from _O_. Then, since _P_ is the middle point of _AB_, _x_ is one half of _OA_ and _y_ is one half of _OB_. The area of the parallelogram whose adjacent sides are _x_ and _y_ is one half the area of the triangle _AOB_, and therefore, by the preceding paragraph, is constant. This area is equal to _xy · __sin__ α_, where α is the constant angle between the asymptotes. It follows that the product _xy_ is constant, and since _x_ and _y_ are the oblique coördinates of the point _P_, the asymptotes being the axes of reference, we have

_The equation of the hyperbola, referred to the asymptotes as axes, is __xy =__ constant._

This identifies the curve with the hyperbola as defined and discussed in works on analytic geometry.

[Figure 30]

FIG. 30

119. Equation of parabola. We have defined the parabola as a conic which is tangent to the line at infinity (§ 110). Draw now two tangents to the curve (Fig. 30), meeting in _A_, the points of contact being _B_ and _C_. These two tangents, together with the line at infinity, form a triangle circumscribed about the conic. Draw through _B_ a parallel to _AC_, and through _C_ a parallel to _AB_. If these meet in _D_, then

_AD_ is a diameter. Let _AD_ meet the curve in _P_, and the chord _BC_ in _Q_. _P_ is then the middle point of _AQ_. Also, _Q_ is the middle point of the chord _BC_, and therefore the diameter _AD_ bisects all chords parallel to _BC_. In particular, _AD_ passes through _P_, the point of contact of the tangent drawn parallel to _BC_.

Draw now another tangent, meeting _AB_ in _B'_ and _AC_ in _C'_. Then these three, with the line at infinity, make a circumscribed quadrilateral. But, by Brianchon's theorem applied to a quadrilateral (§ 88), it appears that a parallel to _AC_ through _B'_, a parallel to _AB_ through _C'_, and the line _BC_ meet in a point _D'_. Also, from the similar triangles _BB'D'_ and _BAC_ we have, for all positions of the tangent line _B'C_,

_B'D' : BB' = AC : AB,_

or, since _B'D' = AC'_,

_AC': BB' = AC:AB =_ constant.

If another tangent meet _AB_ in _B"_ and _AC_ in _C"_, we have

_ AC' : BB' = AC" : BB", _

and by subtraction we get

_C'C" : B'B" =_ constant;

whence

The segments cut off on any two tangents to a parabola by a variable tangent are proportional.

If now we take the tangent _B'C'_ as axis of ordinates, and the diameter through the point of contact _O_ as axis of abscissas, calling the coordinates of _B(x, y)_ and of _C(x', y')_, then, from the similar triangles _BMD'_ and we have

_y : y' = BD' : D'C = BB' : AB'._

Also

_y : y' = B'D' : C'C = AC' : C'C._

If now a line is drawn through _A_ parallel to a diameter, meeting the axis of ordinates in _K_, we have

_AK : OQ' = AC' : CC' = y : y',_

and

_OM : AK = BB' : AB' = y : y',_

and, by multiplication,

_OM : OQ' = y__2__ : y'__2__,_

or

_x : x’ = y__2__ : y’__2__;_

whence

The abscissas of two points on a parabola are to each other as the squares of the corresponding coördinates, a diameter and the tangent to the curve at the extremity of the diameter being the axes of reference.

The last equation may be written

_y__2__ = 2px,_

where _2p_ stands for _y’__2__ : x’_.

The parabola is thus identified with the curve of the same name studied in treatises on analytic geometry.

120. Equation of central conics referred to conjugate diameters. Consider now a _central conic_, that is, one which is not a parabola and the center of which is therefore at a finite distance. Draw any four tangents to it, two of which are parallel (Fig. 31). Let the parallel tangents meet one of the other tangents in _A_ and _B_ and the other in _C_ and _D_, and let _P_ and _Q_ be the points of contact of the parallel tangents _R_ and _S_ of the others. Then _AC_, _BD_, _PQ_, and _RS_ all meet in a point _W_ (§ 88). From the figure,

_PW : WQ = AP : QC = PD : BQ,_

or

_AP · BQ = PD · QC._

If now _DC_ is a fixed tangent and _AB_ a variable one, we have from this equation

_AP · BQ = __constant._

This constant will be positive or negative according as _PA_ and _BQ_ are measured in the same or in opposite directions. Accordingly we write

_AP · BQ = ± b__2__._

[Figure 31]

FIG. 31

Since _AD_ and _BC_ are parallel tangents, _PQ_ is a diameter and the conjugate diameter is parallel to _AD_. The middle point of _PQ_ is the center of the conic. We take now for the axis of abscissas the diameter _PQ_, and the conjugate diameter for the axis of ordinates. Join _A_ to _Q_ and _B_ to _P_ and draw a line through _S_ parallel to the axis of ordinates. These three lines all meet in a point _N_, because _AP_, _BQ_, and _AB_ form a triangle circumscribed to the conic. Let _NS_ meet _PQ_ in _M_. Then, from the properties of the circumscribed triangle (§ 89), _M_, _N_, _S_, and the point at infinity on _NS_ are four harmonic points, and therefore _N_ is the

middle point of _MS_. If the coördinates of _S_ are _(x, y)_, so that _OM_ is _x_ and _MS_ is _y_, then _MN = y/2_. Now from the similar triangles _PMN_ and _PQB_ we have

_BQ : PQ = NM : PM,_

and from the similar triangles _PQA_ and _MQN_,

_AP : PQ = MN : MQ,_

whence, multiplying, we have

_±b__2__/4 a__2__ = y__2__/4 (a + x)(a - x),_

where

[formula]

or, simplifying,

[formula]

which is the equation of an ellipse when _b__2_ has a positive sign, and of a hyperbola when _b__2_ has a negative sign. We have thus identified point-rows of the second order with the curves given by equations of the second degree.

PROBLEMS

1. Draw a chord of a given conic which shall be bisected by a given point _P_.

2. Show that all chords of a given conic that are bisected by a given chord are tangent to a parabola.

3. Construct a parabola, given two tangents with their points of contact.

4. Construct a parabola, given three points and the direction of the diameters.

5. A line _u’_ is drawn through the pole _U_ of a line _u_ and at right angles to _u_. The line _u_ revolves about a point _P_. Show that the line _u’_ is tangent to a parabola. (The lines _u_ and _u’_ are called normal conjugates.)

6. Given a circle and its center _O_, to draw a line through a given point _P_ parallel to a given line _q_. Prove the following construction: Let _p_ be the polar of _P_, _Q_ the pole of _q_, and _A_ the intersection of _p_ with _OQ_. The polar of _A_ is the desired line.

CHAPTER VIII - INVOLUTION

[Figure 32]

FIG. 32

121. Fundamental theorem. The important theorem concerning two complete quadrangles (§ 26), upon which the theory of four harmonic points was based, can easily be extended to the case where the four lines _KL_, _K'L'_, _MN_, _M'N'_ do not all meet in the same point _A_, and the more general theorem that results may also be made the basis of a theory no less important, which has to do with six points on a line. The theorem is as follows:

_Given two complete quadrangles, __K__, __L__, __M__, __N__ and __K'__, __L'__, __M'__, __N'__, so related that __KL__ and __K'L'__ meet in __A__, __MN__ and __M'N'__ in __A'__, __KN__ and __K'N'__ in __B__, __LM__ and __L'M'__ in __B'__, __LN__ and __L'N'__ in __C__, and __KM__ and __K'M'__ in __C'__, then, if __A__, __A'__, __B__, __B'__, and __C__ are in a straight line, the point __C'__ also lies on that straight line._

The theorem follows from Desargues's theorem (Fig. 32). It is seen that _KK'_, _LL'_, _MM'_, _NN'_ all meet in a point, and thus, from the same theorem, applied to the triangles _KLM_ and _K'L'M'_, the point _C'_ is on the same line with _A_ and _B'_. As in the simpler case, it is seen that there is an indefinite number of quadrangles which may be drawn, two sides of which go through _A_ and _A'_, two through _B_ and _B'_, and one through _C_. The sixth side must then go through _C'_. Therefore,

122. _Two pairs of points, __A__, __A'__ and __B__, __B'__, being given, then the point __C'__ corresponding to any given point __C__ is uniquely determined._

The construction of this sixth point is easily accomplished. Draw through _A_ and _A'_ any two lines, and cut across them by any line through _C_ in the points _L_ and _N_. Join _N_ to _B_ and _L_ to _B'_, thus determining the points _K_ and _M_ on the two lines through _A_ and _A'_, The line _KM_ determines the desired point _C'_. Manifestly, starting from _C'_, we come in this way always to the same point _C_. The particular quadrangle employed is of no consequence. Moreover, since one pair of opposite sides in a complete quadrangle is not distinguishable in any way from any other, the same set of six points will be obtained by starting from the pairs _AA'_ and _CC'_, or from the pairs _BB'_ and _CC'_.

123. Definition of involution of points on a line.

Three pairs of points on a line are said to be in involution if through each pair may be drawn a pair of opposite sides of a complete quadrangle. If two pairs are fixed and one of the third pair describes the line, then the other also describes the line, and the points of the line are said to be paired in the involution determined by the two fixed pairs.

[Figure 33]

FIG. 33

124. Double-points in an involution. The points _C_ and _C'_ describe projective point-rows, as may be seen by fixing the points _L_ and _M_. The self-corresponding points, of which there are two or none, are called the _double-points_ in the

involution. It is not difficult to see that the double-points in the involution are harmonic conjugates with respect to corresponding points in the involution. For, fixing as before the points _L_ and _M_, let the intersection of the lines _CL_ and _C'M_ be _P_ (Fig. 33). The locus of _P_ is a conic which goes through the double-points, because the point-rows _C_ and _C'_ are projective, and therefore so are the pencils _LC_ and _MC'_ which generate the locus of _P_. Also, when _C_ and _C'_ fall together, the point _P_ coincides with them. Further, the tangents at _L_ and _M_ to this conic described by _P_ are the lines _LB_ and _MB_. For in the pencil at _L_ the ray _LM_ common to the two pencils which generate the conic is the ray _LB'_ and corresponds to the ray _MB_ of _M_, which is therefore the tangent line to the conic at _M_. Similarly for the tangent _LB_ at _L_. _LM_ is therefore the polar of _B_ with respect to this conic, and _B_ and _B'_ are therefore harmonic conjugates with respect to the double-points. The same discussion applies to any other pair of corresponding points in the involution.

[Figure 34]

FIG. 34

125. Desargues's theorem concerning conics through four points. Let _DD'_ be any pair of points in the involution determined as above, and consider the conic passing through the five points _K_, _L_, _M_, _N_, _D_. We shall use Pascal's theorem to show that this conic also passes through _D'_. The point _D'_ is determined as follows: Fix _L_ and _M_ as before (Fig. 34) and join _D_ to _L_, giving on _MN_ the point _N'_. Join _N'_ to _B_, giving on _LK_ the point _K'_. Then _MK'_ determines the point _D'_ on the line _AA'_, given by the complete quadrangle _K'_, _L_, _M_, _N'_. Consider the following six points, numbering them in order: _D = 1_, _D' = 2_, _M = 3_, _N = 4_, _K = 5_, and _L = 6_. We have the following intersections: _B = (12-45)_, _K' = (23-56)_, _N' = (34-61)_; and since by construction _B_, _N_, and _K'_ are on a straight line, it follows from the converse of Pascal's theorem, which is easily established, that the six points are on a conic. We have, then, the beautiful theorem due to Desargues:

The system of conics through four points meets any line in the plane in pairs of points in involution.

126. It appears also that the six points in involution determined by the quadrangle through the four fixed points belong also to the same involution with the points cut out by the system of conics, as indeed we might infer from the fact that the three pairs of opposite sides of the quadrangle may be considered as degenerate conics of the system.

127. Conics through four points touching a given line. It is further evident that the involution determined on a line by the system of conics will have a double-point where a conic of the system is tangent to the line. We may therefore infer the theorem

Through four fixed points in the plane two conics or none may be drawn tangent to any given line.

[Figure 35]

FIG. 35

128. Double correspondence. We have seen that corresponding points in an involution form two projective point-rows superposed on the same straight line. Two projective point-rows superposed on the same straight line are, however, not necessarily in involution, as a simple example will show. Take two lines, _a_ and _a’_, which both revolve about a fixed point _S_ and which always make the same angle with each other (Fig. 35). These lines cut out on any line in the plane which does not pass through _S_ two projective point-rows, which are not, however, in involution unless the angle between the lines is a right angles. For a point _P_ may correspond to a point _P’_, which in turn will correspond to some other point than _P_. The peculiarity of point-rows in involution is that any point will correspond to the same point, in whichever point-row it is considered as belonging. In this case, if a point _P_ corresponds to a point _P’_, then the point _P’_ corresponds back again to the point _P_. The points _P_ and _P’_ are then said to _correspond doubly_. This notion is worthy of further study.

[Figure 36]

FIG. 36

129. Steiner’s construction. It will be observed that the solution of the fundamental problem given in § 83, _Given three pairs of points of two protective point-rows, to construct other pairs_, cannot be carried out if the two point-rows lie on the same straight line. Of course the method may be easily altered to cover that case also, but it is worth while to give another solution of the problem, due to Steiner, which will also give further information regarding the theory of involution, and which may, indeed, be used as a foundation for that theory. Let the two point-rows _A_, _B_, _C_, _D_, ... and _A’_, _B’_, _C’_, _D’_, ... be superposed on the line _u_. Project them both to a point _S_ and pass any conic _κ_ through _S_. We thus obtain two projective pencils, _a_, _b_, _c_, _d_, ... and _a’_, _b’_, _c’_, _d’_, ... at _S_, which meet the conic in the points _α_, _β_, _γ_, _δ_, ... and _α’_, _β’_, _γ’_, _δ’_, ... (Fig. 36). Take now _γ_ as the center of a pencil projecting the points _α’_, _β’_, _δ’_, ..., and take _γ’_ as the center of a pencil projecting the points _α_, _β_, _δ_, ... These two pencils are projective to each other, and since they have a self-correspondin ray in common, they are in perspective position and corresponding rays meet on the line joining _(γα’, γ’α)_ to _(γβ’, γ’β)_. The correspondence between points in the two point-rows on _u_ is now easily traced.

130. Application of Steiner’s construction to double correspondence. Steiner’s construction throws into our hands an important theorem concerning double correspondence: _If two projective point-rows, superposed on the same line, have one

pair of points which correspond to each other doubly, then all pairs correspond to each other doubly, and the line is paired in involution._ To make this appear, let us call the point _A_ on _u_ by two names, _A_ and _P’_, according as it is thought of as belonging to the one or to the other of the two point-rows. If this point is one of a pair which correspond to each other doubly, then the points _A’_ and _P_ must coincide (Fig. 37). Take now any point _C_, which we will also call _R’_. We must show that the corresponding point _C’_ must also coincide with the point _B_. Join all the points to _S_, as before, and it appears that the points α and _π’_ coincide, as also do the points _α’π_ and _γρ’_. By the above construction the line _γ’ρ_ must meet _γρ’_ on the line joining _(γα’, γ’α)_ with _(γπ’, γ’π)_. But these four points form a quadrangle inscribed in the conic, and we know by § 95 that the tangents at the opposite vertices _γ_ and _γ’_ meet on the line _v_. The line _γ’ρ_ is thus a tangent to the conic, and _C’_ and _R_ are the same point. That two projective point-rows superposed on the same line are also in involution when one pair, and therefore all pairs, correspond doubly may be shown by taking _S_ at one vertex of a complete quadrangle which has two pairs of opposite sides going through two pairs of points. The details we leave to the student.

[Figure 37]

FIG. 37

[Figure 38]

FIG. 38

131. Involution of points on a point-row of the second order. It is important to note also, in Steiner’s construction, that we have obtained two point-rows of the second order superposed on the same conic, and have paired the points of one with the points of the other in such a way that the correspondence is double. We may then extend the notion of involution to point-rows of the second order and say that _the points of a conic are paired in involution when they are corresponding __ points of two projective point-rows superposed on the conic, and when they correspond to each other doubly._ With this definition we may prove the theorem: _The lines joining corresponding points of a point-row of the second order in involution all pass through a fixed point __U__, and the line joining any two points __A__, __B__ meets the line joining the two corresponding points __A’__, __B’__ in the points of a line __u__, which is the polar of __U__ with respect to the conic._ For take _A_ and _A’_ as the centers of two pencils, the first perspective to the point-row _A’_, _B’_, _C’_ and the second perspective to the point-row _A_, _B_, _C_. Then, since the common ray of the two pencils corresponds to itself, they are in perspective position, and their axis of perspectivity _u_ (Fig. 38) is the line which joins the point _(AB’, A’B)_ to the point _(AC’, A’C)_. It is then immediately clear, from the theory of poles and polars, that _BB’_ and _CC’_ pass through the pole _U_ of the line _u_.

132. Involution of rays. The whole theory thus far developed may be dualized, and a theory of lines in involution may be built up, starting with the complete

quadrilateral. Thus,

_The three pairs of rays which may be drawn from a point through the three pairs of opposite vertices of a complete quadrilateral are said to be in involution. If the pairs __aa'__ and __bb'__ are fixed, and the line __c__ describes a pencil, the corresponding line __c'__ also describes a pencil, and the rays of the pencil are said to be paired in the involution determined by __aa'__ and __bb'__._

133. Double rays. The self-corresponding rays, of which there are two or none, are called _double rays_ of the involution. Corresponding rays of the involution are harmonic conjugates with respect to the double rays. To the theorem of Desargues (§ 125) which has to do with the system of conics through four points we have the dual:

The tangents from a fixed point to a system of conics tangent to four fixed lines form a pencil of rays in involution.

134. If a conic of the system should go through the fixed point, it is clear that the two tangents would coincide and indicate a double ray of the involution. The theorem, therefore, follows:

Two conics or none may be drawn through a fixed point to be tangent to four fixed lines.

135. Double correspondence. It further appears that two projective pencils of rays which have the same center are in involution if two pairs of rays correspond to each other doubly. From this it is clear that we might have deemed six rays in involution as six rays which pass through a point and also through six points in involution. While this would have been entirely in accord with the treatment which was given the corresponding problem in the theory of harmonic points and lines, it is more satisfactory, from an aesthetic point of view, to build the theory of lines in involution on its own base. The student can show, by methods entirely analogous to those used in the second chapter, that involution is a projective property; that is, six rays in involution are cut by any transversal in six points in involution.

136. Pencils of rays of the second order in involution. We may also extend the notion of involution to pencils of rays of the second order. Thus, _the tangents to a conic are in involution when they are corresponding rays of two protective pencils of the second order superposed upon the same conic, and when they correspond to each other doubly._ We have then the theorem:

137. _The intersections of corresponding rays of a pencil of the second order in involution are all on a straight line __u__, and the intersection of any two tangents __ab__, when joined to the intersection of the corresponding tangents __a'b'__, gives a line which passes through a fixed point __U__, the pole of the line __u__ with respect to the conic._

138. Involution of rays determined by a conic. We have seen in the theory of poles and polars (§ 103) that if a point _P_ moves along a line _m_, then the polar of _P_ revolves about a point. This pencil cuts out on _m_ another point-row _P'_, projective also to _P_. Since the polar of _P_ passes through _P'_, the polar of _P'_ also passes through _P_, so that the correspondence between _P_ and _P'_ is double. The

two point-rows are therefore in involution, and the double points, if any exist, are the points where the line _m_ meets the conic. A similar involution of rays may be found at any point in the plane, corresponding rays passing each through the pole of the other. We have called such points and rays _conjugate_ with respect to the conic (§ 100). We may then state the following important theorem:

139. _A conic determines on every line in its plane an involution of points, corresponding points in the involution __ being conjugate with respect to the conic. The double points, if any exist, are the points where the line meets the conic._

140. The dual theorem reads: _A conic determines at every point in the plane an involution of rays, corresponding rays being conjugate with respect to the conic. The double rays, if any exist, are the tangents from the point to the conic._

PROBLEMS

1. Two lines are drawn through a point on a conic so as always to make right angles with each other. Show that the lines joining the points where they meet the conic again all pass through a fixed point.

2. Two lines are drawn through a fixed point on a conic so as always to make equal angles with the tangent at that point. Show that the lines joining the two points where the lines meet the conic again all pass through a fixed point.

3. Four lines divide the plane into a certain number of regions. Determine for each region whether two conics or none may be drawn to pass through points of it and also to be tangent to the four lines.

4. If a variable quadrangle move in such a way as always to remain inscribed in a fixed conic, while three of its sides turn each around one of three fixed collinear points, then the fourth will also turn around a fourth fixed point collinear with the other three.

5. State and prove the dual of problem 4.

6. Extend problem 4 as follows: If a variable polygon of an even number of sides move in such a way as always to remain inscribed in a fixed conic, while all its sides but one pass through as many fixed collinear points, then the last side will also pass through a fixed point collinear with the others.

7. If a triangle _QRS_ be inscribed in a conic, and if a transversal _s_ meet two of its sides in _A_ and _A’_, the third side and the tangent at the opposite vertex in _B_ and _B’_, and the conic itself in _C_ and _C’_, then _AA’_, _BB’_, _CC’_ are three pairs of points in an involution.

8. Use the last exercise to solve the problem: Given five points, _Q_, _R_, _S_, _C_, _C’_, on a conic, to draw the tangent at any one of them.

9. State and prove the dual of problem 7 and use it to prove the dual of problem 8.

10. If a transversal cut two tangents to a conic in _B_ and _B’_, their chord of contact in _A_, and the conic itself in _P_ and _P’_, then the point _A_ is a double point

of the involution determined by _BB’_ and _PP’_.

11. State and prove the dual of problem 10.

12. If a variable conic pass through two given points, _P_ and _P’_, and if it be tangent to two given lines, the chord of contact of these two tangents will always pass through a fixed point on _PP’_.

13. Use the last theorem to solve the problem: Given four points, _P_, _P’_, _Q_, _S_, on a conic, and the tangent at one of them, _Q_, to draw the tangent at any one of the other points, _S_.

14. Apply the theorem of problem 9 to the case of a hyperbola where the two tangents are the asymptotes. Show in this way that if a hyperbola and its asymptotes be cut by a transversal, the segments intercepted by the curve and by the asymptotes respectively have the same middle point.

15. In a triangle circumscribed about a conic, any side is divided harmonically by its point of contact and the point where it meets the chord joining the points of contact of the other two sides.

CHAPTER IX - METRICAL PROPERTIES OF INVOLUTIONS

[Figure 39]

FIG. 39

141. Introduction of infinite point; center of involution. We connect the projective theory of involution with the metrical, as usual, by the introduction of the elements at infinity. In an involution of points on a line the point which corresponds to the infinitely distant point is called the _center_ of the involution. Since corresponding points in the involution have been shown to be harmonic conjugates with respect to the double points, the center is midway between the double points when they exist. To construct the center (Fig. 39) we draw as usual through _A_ and _A’_ any two rays and cut them by a line parallel to _AA’_ in the points _K_ and _M_. Join these points to _B_ and _B’_, thus determining on _AK_ and _AN_ the points _L_ and _N_. _LN_ meets _AA’_ in the center _O_ of the involution.

142. Fundamental metrical theorem. From the figure we see that the triangles _OLB’_ and _PLM_ are similar, _P_ being the intersection of KM and LN. Also the triangles _KPN_ and _BON_ are similar. We thus have

$$_OB : PK = ON : PN_$$

and

$$_OB’ : PM = OL : PL;_$$

whence

_OB · OB’ : PK · PM = ON · OL : PN · PL._

In the same way, from the similar triangles _OAL_ and _PKL_, and also _OA’N_ and _PMN_, we obtain

_OA · OA’ : PK · PM = ON · OL : PN · PL,_

and this, with the preceding, gives at once the fundamental theorem, which is sometimes taken also as the definition of involution:

_OA · OA’ = OB · OB’ = __constant__,_

or, in words,

The product of the distances from the center to two corresponding points in an involution of points is constant.

143. Existence of double points. Clearly, according as the constant is positive or negative the involution will or will not have double points. The constant is the square root of the distance from the center to the double points. If _A_ and _A’_ lie both on the same side of the center, the product _OA · OA’_ is positive; and if they lie on opposite sides, it is negative. Take the case where they both lie on the same side of the center, and take also the pair of corresponding points _BB’_. Then, since _OA · OA’ = OB · OB’_, it cannot happen that _B_ and _B’_ are separated from each other by _A_ and _A’_. This is evident enough if the points are on opposite sides of the center. If the pairs are on the same side of the center, and _B_ lies between _A_ and _A’_, so that _OB_ is greater, say, than _OA_, but less than _OA’_, then, by the equation _OA · OA’ = OB · OB’_, we must have _OB’_ also less than _OA’_ and greater than _OA_. A similar discussion may be made for the case where _A_ and _A’_ lie on opposite sides of _O_. The results may be stated as follows, without any reference to the center:

Given two pairs of points in an involution of points, if the points of one pair are separated from each other by the points of the other pair, then the involution has no double points. If the points of one pair are not separated from each other by the points of the other pair, then the involution has two double points.

144. An entirely similar criterion decides whether an involution of rays has or has not double rays, or whether an involution of planes has or has not double planes.

[Figure 40]

FIG. 40

145. Construction of an involution by means of circles. The equation just derived, _OA · OA’ = OB · OB’_, indicates another simple way in which points of an involution of points may be constructed. Through _A_ and _A’_ draw any circle, and draw also any circle through _B_ and _B’_ to cut the first in the two points _G_ and _G’_ (Fig.

40). Then any circle through _G_ and _G’_ will meet the line in pairs of points in the involution determined by _AA’_ and _BB’_. For if such a circle meets the line in the points _CC’_, then, by the theorem in the geometry of the circle which says that _if any chord is __ drawn through a fixed point within a circle, the product of its segments is constant in whatever direction the chord is drawn, and if a secant line be drawn from a fixed point without a circle, the product of the secant and its external segment is constant in whatever direction the secant line is drawn_, we have _OC · OC’ = OG · OG’ =_ constant. So that for all such points _OA · OA’ = OB · OB’ = OC · OC’_. Further, the line _GG’_ meets _AA’_ in the center of the involution. To find the double points, if they exist, we draw a tangent from _O_ to any of the circles through _GG’_. Let _T_ be the point of contact. Then lay off on the line _OA_ a line _OF_ equal to _OT_. Then, since by the above theorem of elementary geometry _OA · OA’ = OT__2__ = OF__2_, we have one double point _F_. The other is at an equal distance on the other side of _O_. This simple and effective method of constructing an involution of points is often taken as the basis for the theory of involution. In projective geometry, however, the circle, which is not a figure that remains unaltered by projection, and is essentially a metrical notion, ought not to be used to build up the purely projective part of the theory.

146. It ought to be mentioned that the theory of analytic geometry indicates that the circle is a special conic section that happens to pass through two particular imaginary points on the line at infinity, called the _circular points_ and usually denoted by _I_ and _J_. The above method of obtaining a point-row in involution is, then, nothing but a special case of the general theorem of the last chapter (§ 125), which asserted that a system of conics through four points will cut any line in the plane in a point-row in involution.

[Figure 41]

FIG. 41

147. Pairs in an involution of rays which are at right angles. Circular involution. In an involution of rays there is no one ray which may be distinguished from all the others as the point at infinity is distinguished from all other points on a line. There is one pair of rays, however, which does differ from all the others in that for this particular pair the angle is a right angle. This is most easily shown by using the construction that employs circles, as indicated above. The centers of all the circles through _G_ and _G’_ lie on the perpendicular bisector of the line _GG’_. Let this line meet the line _AA’_ in the point _C_ (Fig. 41), and draw the circle with center _C_ which goes through _G_ and _G’_. This circle cuts out two points _M_ and _M’_ in the involution. The rays _GM_ and _GM’_ are clearly at right angles, being inscribed in a semicircle. If, therefore, the involution of points is projected to _G_, we have found two corresponding rays which are at right angles to each other. Given now any involution of rays with center _G_, we may cut across it by a straight line and proceed to find the two points _M_ and _M’_. Clearly there will be only one such pair unless the perpendicular bisector of _GG’_ coincides with the line _AA’_. In this case every ray is at right angles to its corresponding ray, and the involution is called _circular_.

148. Axes of conics. At the close of the last chapter (§ 140) we gave the theorem: _A conic determines at every point in its plane an involution of rays, corresponding rays __ being conjugate with respect to the conic. The double rays, if any exist, are the tangents from the point to the conic._ In particular, taking the point as the center of the conic, we find that conjugate diameters form a system of rays in involution, of which the asymptotes, if there are any, are the double rays. Also, conjugate diameters are harmonic conjugates with respect to the asymptotes. By the theorem of the last paragraph, there are two conjugate diameters which are at right angles to each other. These are called axes. In the case of the parabola, where the center is at infinity, and on the curve, there are, properly speaking, no conjugate diameters. While the line at infinity might be considered as conjugate to all the other diameters, it is not possible to assign to it any particular direction, and so it cannot be used for the purpose of defining an axis of a parabola. There is one diameter, however, which is at right angles to its conjugate system of chords, and this one is called the _axis_ of the parabola. The circle also furnishes an exception in that every diameter is an axis. The involution in this case is circular, every ray being at right angles to its conjugate ray at the center.

149. Points at which the involution determined by a conic is circular. It is an important problem to discover whether for any conic other than the circle it is possible to find any point in the plane where the involution determined as above by the conic is circular. We shall proceed to the curious problem of proving the existence of such points and of determining their number and situation. We shall then develop the important properties of such points.

150. It is clear, in the first place, that such a point cannot be on the outside of the conic, else the involution would have double rays and such rays would have to be at right angles to themselves. In the second place, if two such points exist, the line joining them must be a diameter and, indeed, an axis. For if _F_ and _F’_ were two such points, then, since the conjugate ray at _F_ to the line _FF’_ must be at right angles to it, and also since the conjugate ray at _F’_ to the line _FF’_ must be at right angles to it, the pole of _FF’_ must be at infinity in a direction at right angles to _FF’_. The line _FF’_ is then a diameter, and since it is at right angles to its conjugate diameter, it must be an axis. From this it follows also that the points we are seeking must all lie on one of the two axes, else we should have a diameter which does not go through the intersection of all axes—the center of the conic. At least one axis, therefore, must be free from any such points.

[Figure 42]

FIG. 42

151. Let now _P_ be a point on one of the axes (Fig. 42), and draw any ray through it, such as _q_. As _q_ revolves about _P_, its pole _Q_ moves along a line at right angles to the axis on which _P_ lies, describing a point-row _p_ projective to the pencil of rays _q_. The point at infinity in a direction at right angles to _q_ also describes a point-row projective to _q_. The line joining corresponding points of these

two point-rows is always a conjugate line to _q_ and at right angles to _q_, or, as we may call it, a _conjugate normal_ to _q_. These conjugate normals to _q_, joining as they do corresponding points in two projective point-rows, form a pencil of rays of the second order. But since the point at infinity on the point-row _Q_ corresponds to the point at infinity in a direction at right angles to _q_, these point-rows are in perspective position and the normal conjugates of all the lines through _P_ meet in a point. This point lies on the same axis with _P_, as is seen by taking _q_ at right angles to the axis on which _P_ lies. The center of this pencil may be called _P'_, and thus we have paired the point _P_ with the point _P'_. By moving the point _P_ along the axis, and by keeping the ray _q_ parallel to a fixed direction, we may see that the point-row _P_ and the point-row _P'_ are projective. Also the correspondence is double, and by starting from the point _P'_ we arrive at the point _P_. Therefore the point-rows _P_ and _P'_ are in involution, and if only the involution has double points, we shall have found in them the points we are seeking. For it is clear that the rays through _P_ and the corresponding rays through _P'_ are conjugate normals; and if _P_ and _P'_ coincide, we shall have a point where all rays are at right angles to their conjugates. We shall now show that the involution thus obtained on one of the two axes must have double points.

[Figure 43]

FIG. 43

152. Discovery of the foci of the conic. We know that on one axis no such points as we are seeking can lie (§ 150). The involution of points _PP'_ on this axis can therefore have no double points. Nevertheless, let _PP'_ and _RR'_ be two pairs of corresponding points on this axis (Fig. 43). Then we know that _P_ and _P'_ are separated from each other by _R_ and _R'_ (§ 143). Draw a circle on _PP'_ as a diameter, and one on _RR'_ as a diameter. These must intersect in two points, _F_ and _F'_, and since the center of the conic is the center of the involution _PP'_, _RR'_, as is easily seen, it follows that _F_ and _F'_ are on the other axis of the conic. Moreover, _FR_ and _FR'_ are conjugate normal rays, since _RFR'_ is inscribed in a semicircle, and the two rays go one through _R_ and the other through _R'_. The involution of points _PP'_, _RR'_ therefore projects to the two points _F_ and _F'_ in two pencils of rays in involution which have for corresponding rays conjugate normals to the conic. We may, then, say:

There are two and only two points of the plane where the involution determined by the conic is circular. These two points lie on one of the axes, at equal distances from the center, on the inside of the conic. These points are called the foci of the conic.

153. The circle and the parabola. The above discussion applies only to the central conics, apart from the circle. In the circle the two foci fall together at the center. In the case of the parabola, that part of the investigation which proves the existence of two foci on one of the axes will not hold, as we have but one axis. It is seen, however, that as _P_ moves to infinity, carrying the line _q_ with it, _q_ becomes

the line at infinity, which for the parabola is a tangent line. Its pole _Q_ is thus at infinity and also the point _P'_, so that _P_ and _P'_ fall together at infinity, and therefore one focus of the parabola is at infinity. There must therefore be another, so that

A parabola has one and only one focus in the finite part of the plane.

[Figure 44]

FIG. 44

154. Focal properties of conics. We proceed to develop some theorems which will exhibit the importance of these points in the theory of the conic section. Draw a tangent to the conic, and also the normal at the point of contact _P_. These two lines are clearly conjugate normals. The two points _T_ and _N_, therefore, where they meet the axis which contains the foci, are corresponding points in the involution considered above, and are therefore harmonic conjugates with respect to the foci (Fig. 44); and if we join them to the point _P_, we shall obtain four harmonic lines. But two of them are at right angles to each other, and so the others make equal angles with them (Problem 4, Chapter II). Therefore

The lines joining a point on the conic to the foci make equal angles with the tangent.

It follows that rays from a source of light at one focus are reflected by an ellipse to the other.

155. In the case of the parabola, where one of the foci must be considered to be at infinity in the direction of the diameter, we have

[Figure 45]

FIG. 45

A diameter makes the same angle with the tangent at its extremity as that tangent does with the line from its point of contact to the focus (Fig. 45).

156. This last theorem is the basis for the construction of the parabolic reflector. A ray of light from the focus is reflected from such a reflector in a direction parallel to the axis of the reflector.

157. Directrix. Principal axis. Vertex. The polar of the focus with respect to the conic is called the _directrix_. The axis which contains the foci is called the _principal axis_, and the intersection of the axis with the curve is called the _vertex_ of the curve. The directrix is at right angles to the principal axis. In a parabola the vertex is equally distant from the focus and the directrix, these three points and the point at infinity on the axis being four harmonic points. In the ellipse the vertex is nearer to the focus than it is to the directrix, for the same reason, and in the hyperbola it is farther from the focus than it is from the directrix.

[Figure 46]

FIG. 46

158. Another definition of a conic. Let _P_ be any point on the directrix through which a line is drawn meeting the conic in the points _A_ and _B_ (Fig. 46). Let the tangents at _A_ and _B_ meet in _T_, and call the focus _F_. Then _TF_ and _PF_ are conjugate lines, and as they pass through a focus they must be at right angles to each other. Let _TF_ meet _AB_ in _C_. Then _P_, _A_, _C_, _B_ are four harmonic points. Project these four points parallel to _TF_ upon the directrix, and we then get the four harmonic points _P_, _M_, _Q_, _N_. Since, now, _TFP_ is a right angle, the angles _MFQ_ and _NFQ_ are equal, as well as the angles _AFC_ and _BFC_. Therefore the triangles _MAF_ and _NFB_ are similar, and _FA : FM = FB : BN_. Dropping perpendiculars _AA_ and _BB'_ upon the directrix, this becomes _FA : AA' = FB : BB'_. We have thus the property often taken as the definition of a conic:

The ratio of the distances from a point on the conic to the focus and the directrix is constant.

[Figure 47]

FIG. 47

159. Eccentricity. By taking the point at the vertex of the conic, we note that this ratio is less than unity for the ellipse, greater than unity for the hyperbola, and equal to unity for the parabola. This ratio is called the _eccentricity_.

[Figure 48]

FIG. 48

160. Sum or difference of focal distances. The ellipse and the hyperbola have two foci and two directrices. The eccentricity, of course, is the same for one focus as for the other, since the curve is symmetrical with respect to both. If the distances from a point on a conic to the two foci are _r_ and _r'_, and the distances from the same point to the corresponding directrices are _d_ and _d'_ (Fig. 47), we have _r : d = r' : d'_; _(r ± r') : (d ± d')_. In the ellipse _(d + d')_ is constant, being the distance between the directrices. In the hyperbola this distance is _(d - d')_. It follows (Fig. 48) that

In the ellipse the sum of the focal distances of any point on the curve is constant, and in the hyperbola the difference between the focal distances is constant.

PROBLEMS

1. Construct the axis of a parabola, given four tangents.

2. Given two conjugate lines at right angles to each other, and let them meet the axis which has no foci on it in the points _A_ and _B_. The circle on _AB_ as diameter will pass through the foci of the conic.

3. Given the axes of a conic in position, and also a tangent with its point of contact, to construct the foci and determine the length of the axes.

4. Given the tangent at the vertex of a parabola, and two other tangents, to find the focus.

5. The locus of the center of a circle touching two given circles is a conic with the centers of the given circles for its foci.

6. Given the axis of a parabola and a tangent, with its point of contact, to find the focus.

7. The locus of the center of a circle which touches a given line and a given circle consists of two parabolas.

8. Let _F_ and _F'_ be the foci of an ellipse, and _P_ any point on it. Produce _PF_ to _G_, making _PG_ equal to _PF'_. Find the locus of _G_.

9. If the points _G_ of a circle be folded over upon a point _F_, the creases will all be tangent to a conic. If _F_ is within the circle, the conic will be an ellipse; if _F_ is without the circle, the conic will be a hyperbola.

10. If the points _G_ in the last example be taken on a straight line, the locus is a parabola.

11. Find the foci and the length of the principal axis of the conics in problems 9 and 10.

12. In problem 10 a correspondence is set up between straight lines and parabolas. As there is a fourfold infinity of parabolas in the plane, and only a twofold infinity of straight lines, there must be some restriction on the parabolas obtained by this method. Find and explain this restriction.

13. State and explain the similar problem for problem 9.

14. The last four problems are a study of the consequences of the following transformation: A point _O_ is fixed in the plane. Then to any point _P_ is made to correspond the line _p_ at right angles to _OP_ and bisecting it. In this correspondence, what happens to _p_ when _P_ moves along a straight line? What corresponds to the theorem that two lines have only one point in common? What to the theorem that the angle sum of a triangle is two right angles? Etc.

CHAPTER X - ON THE HISTORY OF SYNTHETIC PROJECTIVE GEOMETRY

161. Ancient results. The theory of synthetic projective geometry as we have built it up in this course is less than a century old. This is not to say that many of the theorems and principles involved were not discovered much earlier, but isolated theorems do not make a theory, any more than a pile of bricks makes a building. The materials for our building have been contributed by many different workmen from the days of Euclid down to the present time. Thus, the notion of four harmonic points was familiar to the ancients, who considered it from the metrical point of view as the

division of a line internally and externally in the same ratio(1) the involution of six points cut out by any transversal which intersects the sides of a complete quadrilateral as studied by Pappus(2); but these notions were not made the foundation for any general theory. Taken by themselves, they are of small consequence; it is their relation to other theorems and sets of theorems that gives them their importance. The ancients were doubtless familiar with the theorem, _Two lines determine a point, and two points determine a line_, but they had no glimpse of the wonderful law of duality, of which this theorem is a simple example. The principle of projection, by which many properties of the conic sections may be inferred from corresponding properties of the circle which forms the base of the cone from which they are cut—a principle so natural to modern mathematicians—seems not to have occurred to the Greeks. The ellipse, the hyperbola, and the parabola were to them entirely different curves, to be treated separately with methods appropriate to each. Thus the focus of the ellipse was discovered some five hundred years before the focus of the parabola! It was not till 1522 that Verner(3) of Nürnberg undertook to demonstrate the properties of the conic sections by means of the circle.

162. Unifying principles. In the early years of the seventeenth century—that wonderful epoch in the history of the world which produced a Galileo, a Kepler, a Tycho Brahe, a Descartes, a Desargues, a Pascal, a Cavalieri, a Wallis, a Fermat, a Huygens, a Bacon, a Napier, and a goodly array of lesser lights, to say nothing of a Rembrandt or of a Shakespeare—there began to appear certain unifying principles connecting the great mass of material dug out by the ancients. Thus, in 1604 the great astronomer Kepler(4) introduced the notion that parallel lines should be considered as meeting at an infinite distance, and that a parabola is at once the limiting case of an ellipse and of a hyperbola. He also attributes to the parabola a "blind focus" (_caecus focus_) at infinity on the axis.

163. Desargues. In 1639 Desargues,(5) an architect of Lyons, published a little treatise on the conic sections, in which appears the theorem upon which we have founded the theory of four harmonic points (§ 25). Desargues, however, does not make use of it for that purpose. Four harmonic points are for him a special case of six points in involution when two of the three pairs coincide giving double points. His development of the theory of involution is also different from the purely geometric one which we have adopted, and is based on the theorem (§ 142) that the product of the distances of two conjugate points from the center is constant. He also proves the projective character of an involution of points by showing that when six lines pass through a point and through six points in involution, then any transversal must meet them in six points which are also in involution.

164. Poles and polars. In this little treatise is also contained the theory of poles and polars. The polar line is called a _traversal_.(6) The harmonic properties of poles and polars are given, but Desargues seems not to have arrived at the metrical properties which result when the infinite elements of the plane are introduced. Thus he says, "When the _traversal_ is at an infinite distance, all is unimaginable."

165. Desargues's theorem concerning conics through four points. We find in this little book the beautiful theorem concerning a quadrilateral inscribed in a conic

section, which is given by his name in § 138. The theorem is not given in terms of a system of conics through four points, for Desargues had no conception of any such system. He states the theorem, in effect, as follows: _Given a simple quadrilateral inscribed in a conic section, every transversal meets the conic and the four sides of the quadrilateral in six points which are in involution._

166. Extension of the theory of poles and polars to space. As an illustration of his remarkable powers of generalization, we may note that Desargues extended the notion of poles and polars to space of three dimensions for the sphere and for certain other surfaces of the second degree. This is a matter which has not been touched on in this book, but the notion is not difficult to grasp. If we draw through any point _P_ in space a line to cut a sphere in two points, _A_ and _S_, and then construct the fourth harmonic of _P_ with respect to _A_ and _B_, the locus of this fourth harmonic, for various lines through _P_, is a plane called the _polar plane_ of _P_ with respect to the sphere. With this definition and theorem one can easily find dual relations between points and planes in space analogous to those between points and lines in a plane. Desargues closes his discussion of this matter with the remark, "Similar properties may be found for those other solids which are related to the sphere in the same way that the conic section is to the circle." It should not be inferred from this remark, however, that he was acquainted with all the different varieties of surfaces of the second order. The ancients were well acquainted with the surfaces obtained by revolving an ellipse or a parabola about an axis. Even the hyperboloid of two sheets, obtained by revolving the hyperbola about its major axis, was known to them, but probably not the hyperboloid of one sheet, which results from revolving a hyperbola about the other axis. All the other solids of the second degree were probably unknown until their discovery by Euler.(7)

167. Desargues had no conception of the conic section of the locus of intersection of corresponding rays of two projective pencils of rays. He seems to have tried to describe the curve by means of a pair of compasses, moving one leg back and forth along a straight line instead of holding it fixed as in drawing a circle. He does not attempt to define the law of the movement necessary to obtain a conic by this means.

168. Reception of Desargues’s work. Strange to say, Desargues’s immortal work was heaped with the most violent abuse and held up to ridicule and scorn! "Incredible errors! Enormous mistakes and falsities! Really it is impossible for anyone who is familiar with the science concerning which he wishes to retail his thoughts, to keep from laughing!" Such were the comments of reviewers and critics. Nor were his detractors altogether ignorant and uninstructed men. In spite of the devotion of his pupils and in spite of the admiration and friendship of men like Descartes, Fermat, Mersenne, and Roberval, his book disappeared so completely that two centuries after the date of its publication, when the French geometer Chasles wrote his history of geometry, there was no means of estimating the value of the work done by Desargues. Six years later, however, in 1845, Chasles found a manuscript copy of the "Bruillon-project," made by Desargues’s pupil, De la Hire.

169. Conservatism in Desargues's time. It is not necessary to suppose that this effacement of Desargues's work for two centuries was due to the savage attacks of his critics. All this was in accordance with the fashion of the time, and no man escaped bitter denunciation who attempted to improve on the methods of the ancients. Those were days when men refused to believe that a heavy body falls at the same rate as a lighter one, even when Galileo made them see it with their own eyes at the foot of the tower of Pisa. Could they not turn to the exact page and line of Aristotle which declared that the heavier body must fall the faster! "I have read Aristotle's writings from end to end, many times," wrote a Jesuit provincial to the mathematician and astronomer, Christoph Scheiner, at Ingolstadt, whose telescope seemed to reveal certain mysterious spots on the sun, "and I can assure you I have nowhere found anything similar to what you describe. Go, my son, and tranquilize yourself; be assured that what you take for spots on the sun are the faults of your glasses, or of your eyes." The dead hand of Aristotle barred the advance in every department of research. Physicians would have nothing to do with Harvey's discoveries about the circulation of the blood. "Nature is accused of tolerating a vacuum!" exclaimed a priest when Pascal began his experiments on the Puy-de-Dome to show that the column of mercury in a glass tube varied in height with the pressure of the atmosphere.

170. Desargues's style of writing. Nevertheless, authority counted for less at this time in Paris than it did in Italy, and the tragedy enacted in Rome when Galileo was forced to deny his inmost convictions at the bidding of a brutal Inquisition could not have been staged in France. Moreover, in the little company of scientists of which Desargues was a member the utmost liberty of thought and expression was maintained. One very good reason for the disappearance of the work of Desargues is to be found in his style of writing. He failed to heed the very good advice given him in a letter from his warm admirer Descartes.(8) "You may have two designs, both very good and very laudable, but which do not require the same method of procedure: The one is to write for the learned, and show them some new properties of the conic sections which they do not already know; and the other is to write for the curious unlearned, and to do it so that this matter which until now has been understood by only a very few, and which is nevertheless very useful for perspective, for painting, architecture, etc., shall become common and easy to all who wish to study them in your book. If you have the first idea, then it seems to me that it is necessary to avoid using new terms; for the learned are already accustomed to using those of Apollonius, and will not readily change them for others, though better, and thus yours will serve only to render your demonstrations more difficult, and to turn away your readers from your book. If you have the second plan in mind, it is certain that your terms, which are French, and conceived with spirit and grace, will be better received by persons not preoccupied with those of the ancients.... But, if you have that intention, you should make of it a great volume; explain it all so fully and so distinctly that those gentlemen who cannot study without yawning; who cannot distress their imaginations enough to grasp a proposition in geometry, nor turn the leaves of a book to look at the letters in a figure, shall find nothing in your discourse more difficult to understand than the description of an enchanted palace in a fairy story." The point of

these remarks is apparent when we note that Desargues introduced some seventy new terms in his little book, of which only one, _involution_, has survived. Curiously enough, this is the one term singled out for the sharpest criticism and ridicule by his reviewer, De Beaugrand.(9) That Descartes knew the character of Desargues's audience better than he did is also evidenced by the fact that De Beaugrand exhausted his patience in reading the first ten pages of the book.

171. Lack of appreciation of Desargues. Desargues's methods, entirely different from the analytic methods just then being developed by Descartes and Fermat, seem to have been little understood. "Between you and me," wrote Descartes(10) to Mersenne, "I can hardly form an idea of what he may have written concerning conics." Desargues seems to have boasted that he owed nothing to any man, and that all his results had come from his own mind. His favorite pupil, De la Hire, did not realize the extraordinary simplicity and generality of his work. It is a remarkable fact that the only one of all his associates to understand and appreciate the methods of Desargues should be a lad of sixteen years!

172. Pascal and his theorem. One does not have to believe all the marvelous stories of Pascal's admiring sisters to credit him with wonderful precocity. We have the fact that in 1640, when he was sixteen years old, he published a little placard, or poster, entitled "Essay pour les conique,"(11) in which his great theorem appears for the first time. His manner of putting it may be a little puzzling to one who has only seen it in the form given in this book, and it may be worth while for the student to compare the two methods of stating it. It is given as follows: _"If in the plane of __M__, __S__, __Q__ we draw through __M__ the two lines __MK__ and __MV__, and through the point __S__ the two lines __SK__ and __SV__, and let __K__ be the intersection of __MK__ and __SK__; __V__ the intersection of __MV__ and __SV__; __A__ the intersection of __MA__ and __SA__ (__A__ is the intersection of __SV__ and __MK__), and __μ__ the intersection of __MV__ and __SK__; and if through two of the four points __A__, __K__, __μ__, __V__, which are not in the same straight line with __M__ and __S__, such as __K__ and __V__, we pass the circumference of a circle cutting the lines __MV__, __MP__, __SV__, __SK__ in the points __O__, __P__, __Q__, __N__; I say that the lines __MS__, __NO__, __PQ__ are of the same order."_ (By "lines of the same order" Pascal means lines which meet in the same point or are parallel.) By projecting the figure thus described upon another plane he is able to state his theorem for the case where the circle is replaced by any conic section.

173. It must be understood that the "Essay" was only a résumé of a more extended treatise on conics which, owing partly to Pascal's extreme youth, partly to the difficulty of publishing scientific works in those days, and also to his later morbid interest in religious matters, was never published. Leibniz(12) examined a copy of the complete work, and has reported that the great theorem on the mystic hexagram was made the basis of the whole theory, and that Pascal had deduced some four hundred corollaries from it. This would indicate that here was a man able to take the unconnected materials of projective geometry and shape them into some such symmetrical edifice as we have to-day. Unfortunately for science, Pascal's early death prevented the further development of the subject at his hands.

174. In the "Essay" Pascal gives full credit to Desargues, saying of one of the other propositions, "We prove this property also, the original discoverer of which is M. Desargues, of Lyons, one of the greatest minds of this age ... and I wish to acknowledge that I owe to him the little which I have discovered." This acknowledgment led Descartes to believe that Pascal's theorem should also be credited to Desargues. But in the scientific club which the young Pascal attended in company with his father, who was also a scientist of some reputation, the theorem went by the name of 'la Pascalia,' and Descartes's remarks do not seem to have been taken seriously, which indeed is not to be wondered at, seeing that he was in the habit of giving scant credit to the work of other scientific investigators than himself.

175. De la Hire and his work. De la Hire added little to the development of the subject, but he did put into print much of what Desargues had already worked out, not fully realizing, perhaps, how much was his own and how much he owed to his teacher. Writing in 1679, he says,(13) "I have just read for the first time M. Desargues's little treatise, and have made a copy of it in order to have a more perfect knowledge of it." It was this copy that saved the work of his master from oblivion. De la Hire should be credited, among other things, with the invention of a method by which figures in the plane may be transformed into others of the same order. His method is extremely interesting, and will serve as an exercise for the student in synthetic projective geometry. It is as follows: _Draw two parallel lines, __a__ and __b__, and select a point __P__ in their plane. Through any point __M__ of the plane draw a line meeting __a__ in __A__ and __b__ in __B__. Draw a line through __B__ parallel to __AP__, and let it meet __MP__ in the point __M'__. It may be shown that the point __M'__ thus obtained does not depend at all on the particular ray __MAB__ used in determining it, so that we have set up a one-to-one correspondence between the points __M__ and __M'__ in the plane._ The student may show that as _M_ describes a point-row, _M'_ describes a point-row projective to it. As _M_ describes a conic, _M'_ describes another conic. This sort of correspondence is called a _collineation_. It will be found that the points on the line _b_ transform into themselves, as does also the single point _P_. Points on the line _a_ transform into points on the line at infinity. The student should remove the metrical features of the construction and take, instead of two parallel lines _a_ and _b_, any two lines which may meet in a finite part of the plane. The collineation is a special one in that the general one has an invariant triangle instead of an invariant point and line.

176. Descartes and his influence. The history of synthetic projective geometry has little to do with the work of the great philosopher Descartes, except in an indirect way. The method of algebraic analysis invented by him, and the differential and integral calculus which developed from it, attracted all the interest of the mathematical world for nearly two centuries after Desargues, and synthetic geometry received scant attention during the rest of the seventeenth century and for the greater part of the eighteenth century. It is difficult for moderns to conceive of the richness and variety of the problems which confronted the first workers in the calculus. To come into the possession of a method which would solve almost automatically problems which had baffled the keenest minds of antiquity; to be able to derive in a few moments results which an Archimedes had toiled long and patiently

to reach or a Galileo had determined experimentally; such was the happy experience of mathematicians for a century and a half after Descartes, and it is not to be wondered at that along with this enthusiastic pursuit of new theorems in analysis should come a species of contempt for the methods of the ancients, so that in his preface to his "Méchanique Analytique," published in 1788, Lagrange boasts, "One will find no figures in this work." But at the close of the eighteenth century the field opened up to research by the invention of the calculus began to appear so thoroughly explored that new methods and new objects of investigation began to attract attention. Lagrange himself, in his later years, turned in weariness from analysis and mechanics, and applied himself to chemistry, physics, and philosophical speculations. "This state of mind," says Darboux,(14) "we find almost always at certain moments in the lives of the greatest scholars." At any rate, after lying fallow for almost two centuries, the field of pure geometry was attacked with almost religious enthusiasm.

177. Newton and Maclaurin. But in hastening on to the epoch of Poncelet and Steiner we should not omit to mention the work of Newton and Maclaurin. Although their results were obtained by analysis for the most part, nevertheless they have given us theorems which fall naturally into the domain of synthetic projective geometry. Thus Newton's "organic method"(15) of generating conic sections is closely related to the method which we have made use of in Chapter III. It is as follows: _If two angles, __AOS__ and __AO'S__, of given magnitudes turn about their respective vertices, __O__ and __O'__, in such a way that the point of intersection, __S__, of one pair of arms always lies on a straight line, the point of intersection, __A__, of the other pair of arms will describe a conic._ The proof of this is left to the student.

178. Another method of generating a conic is due to Maclaurin.(16) The construction, which we also leave for the student to justify, is as follows: _If a triangle __C'PQ__ move in such a way that its sides, __PQ__, __QC'__, and __C'P__, turn __ around three fixed points, __R__, __A__, __B__, respectively, while two of its vertices, __P__, __Q__, slide along two fixed lines, __CB'__ and __CA'__, respectively, then the remaining vertex will describe a conic._

179. Descriptive geometry and the second revival. The second revival of pure geometry was again to take place at a time of great intellectual activity. The period at the close of the eighteenth and the beginning of the nineteenth century is adorned with a glorious list of mighty names, among which are Gauss, Lagrange, Legendre, Laplace, Monge, Carnot, Poncelet, Cauchy, Fourier, Steiner, Von Staudt, Möbius, Abel, and many others. The renaissance may be said to date from the invention by Monge(17) of the theory of _descriptive geometry_. Descriptive geometry is concerned with the representation of figures in space of three dimensions by means of space of two dimensions. The method commonly used consists in projecting the space figure on two planes (a vertical and a horizontal plane being most convenient), the projections being made most simply for metrical purposes from infinity in directions perpendicular to the two planes of projection. These two planes are then made to coincide by revolving the horizontal into the vertical about their common line. Such is the method of descriptive geometry which in the hands of Monge

acquired wonderful generality and elegance. Problems concerning fortifications were worked so quickly by this method that the commandant at the military school at Mézières, where Monge was a draftsman and pupil, viewed the results with distrust. Monge afterward became professor of mathematics at Mézières and gathered around him a group of students destined to have a share in the advancement of pure geometry. Among these were Hachette, Brianchon, Dupin, Chasles, Poncelet, and many others.

180. Duality, homology, continuity, contingent relations. Analytic geometry had left little to do in the way of discovery of new material, and the mathematical world was ready for the construction of the edifice. The activities of the group of men that followed Monge were directed toward this end, and we now begin to hear of the great unifying notions of duality, homology, continuity, contingent relations, and the like. The devotees of pure geometry were beginning to feel the need of a basis for their science which should be at once as general and as rigorous as that of the analysts. Their dream was the building up of a system of geometry which should be independent of analysis. Monge, and after him Poncelet, spent much thought on the so-called "principle of continuity," afterwards discussed by Chasles under the name of the "principle of contingent relations." To get a clear idea of this principle, consider a theorem in geometry in the proof of which certain auxiliary elements are employed. These elements do not appear in the statement of the theorem, and the theorem might possibly be proved without them. In drawing the figure for the proof of the theorem, however, some of these elements may not appear, or, as the analyst would say, they become imaginary. "No matter," says the principle of contingent relations, "the theorem is true, and the proof is valid whether the elements used in the proof are real or imaginary."

181. Poncelet and Cauchy. The efforts of Poncelet to compel the acceptance of this principle independent of analysis resulted in a bitter and perhaps fruitless controversy between him and the great analyst Cauchy. In his review of Poncelet’s great work on the projective properties of figures(18) Cauchy says, "In his preliminary discourse the author insists once more on the necessity of admitting into geometry what he calls the ’principle of continuity.’ We have already discussed that principle ... and we have found that that principle is, properly speaking, only a strong induction, which cannot be indiscriminately applied to all sorts of questions in geometry, nor even in analysis. The reasons which we have given as the basis of our opinion are not affected by the considerations which the author has developed in his Traité des Propriétés Projectives des Figures." Although this principle is constantly made use of at the present day in all sorts of investigations, careful geometricians are in agreement with Cauchy in this matter, and use it only as a convenient working tool for purposes of exploration. The one-to-one correspondence between geometric forms and algebraic analysis is subject to many and important exceptions. The field of analysis is much more general than the field of geometry, and while there may be a clear notion in analysis to, correspond to every notion in geometry, the opposite is not true. Thus, in analysis we can deal with four coördinates as well as with three, but the existence of a space of four dimensions to correspond to it does not therefore follow. When the geometer speaks of the two real or imaginary intersections of a straight

line with a conic, he is really speaking the language of algebra. _Apart from the algebra involved_, it is the height of absurdity to try to distinguish between the two points in which a line _fails to meet a conic!_

182. The work of Poncelet. But Poncelet's right to the title "The Father of Modern Geometry" does not stand or fall with the principle of contingent relations. In spite of the fact that he considered this principle the most important of all his discoveries, his reputation rests on more solid foundations. He was the first to study figures _in homology_, which is, in effect, the collineation described in § 175, where corresponding points lie on straight lines through a fixed point. He was the first to give, by means of the theory of poles and polars, a transformation by which an element is transformed into another of a different sort. Point-to-point transformations will sometimes generalize a theorem, but the transformation discovered by Poncelet may throw a theorem into one of an entirely different aspect. The principle of duality, first stated in definite form by Gergonne,(19) the editor of the mathematical journal in which Poncelet published his researches, was based by Poncelet on his theory of poles and polars. He also put into definite form the notions of the infinitely distant elements in space as all lying on a plane at infinity.

183. The debt which analytic geometry owes to synthetic geometry. The reaction of pure geometry on analytic geometry is clearly seen in the development of the notion of the _class_ of a curve, which is the number of tangents that may be drawn from a point in a plane to a given curve lying in that plane. If a point moves along a conic, it is easy to show—and the student is recommended to furnish the proof—that the polar line with respect to a conic remains tangent to another conic. This may be expressed by the statement that the conic is of the second order and also of the second class. It might be thought that if a point moved along a cubic curve, its polar line with respect to a conic would remain tangent to another cubic curve. This is not the case, however, and the investigations of Poncelet and others to determine the class of a given curve were afterward completed by Plücker. The notion of geometrical transformation led also to the very important developments in the theory of invariants, which, geometrically, are the elements and configurations which are not affected by the transformation. The anharmonic ratio of four points is such an invariant, since it remains unaltered under all projective transformations.

184. Steiner and his work. In the work of Poncelet and his contemporaries, Chasles, Brianchon, Hachette, Dupin, Gergonne, and others, the anharmonic ratio enjoyed a fundamental rôle. It is made also the basis of the great work of Steiner,(20) who was the first to treat of the conic, not as the projection of a circle, but as the locus of intersection of corresponding rays of two projective pencils. Steiner not only related to each other, in one-to-one correspondence, point-rows and pencils and all the other fundamental forms, but he set into correspondence even curves and surfaces of higher degrees. This new and fertile conception gave him an easy and direct route into the most abstract and difficult regions of pure geometry. Much of his work was given without any indication of the methods by which he had arrived at it, and many of his results have only recently been verified.

185. Von Staudt and his work. To complete the theory of geometry as we have it to-day it only remained to free it from its dependence on the semimetrical basis of the anharmonic ratio. This work was accomplished by Von Staudt,(21) who applied himself to the restatement of the theory of geometry in a form independent of analytic and metrical notions. The method which has been used in Chapter II to develop the notion of four harmonic points by means of the complete quadrilateral is due to Von Staudt. His work is characterized by a most remarkable generality, in that he is able to discuss real and imaginary forms with equal ease. Thus he assumes a one-to-one correspondence between the points and lines of a plane, and defines a conic as the locus of points which lie on their corresponding lines, and a pencil of rays of the second order as the system of lines which pass through their corresponding points. The point-row and pencil of the second order may be real or imaginary, but his theorems still apply. An illustration of a correspondence of this sort, where the conic is imaginary, is given in § 15 of the first chapter. In defining conjugate imaginary points on a line, Von Staudt made use of an involution of points having no double points. His methods, while elegant and powerful, are hardly adapted to an elementary course, but Reye(22) and others have done much toward simplifying his presentation.

186. Recent developments. It would be only confusing to the student to attempt to trace here the later developments of the science of protective geometry. It is concerned for the most part with curves and surfaces of a higher degree than the second. Purely synthetic methods have been used with marked success in the study of the straight line in space. The struggle between analysis and pure geometry has long since come to an end. Each has its distinct advantages, and the mathematician who cultivates one at the expense of the other will never attain the results that he would attain if both methods were equally ready to his hand. Pure geometry has to its credit some of the finest discoveries in mathematics, and need not apologize for having been born. The day of its usefulness has not passed with the invention of abridged notation and of short methods in analysis. While we may be certain that any geometrical problem may always be stated in analytic form, it does not follow that that statement will be simple or easily interpreted. For many mathematicians the geometric intuitions are weak, and for such the method will have little attraction. On the other hand, there will always be those for whom the subject will have a peculiar glamor—who will follow with delight the curious and unexpected relations between the forms of space. There is a corresponding pleasure, doubtless, for the analyst in tracing the marvelous connections between the various fields in which he wanders, and it is as absurd to shut one’s eyes to the beauties in one as it is to ignore those in the other. "Let us cultivate geometry, then," says Darboux,(23) "without wishing in all points to equal it to its rival. Besides, if we were tempted to neglect it, it would not be long in finding in the applications of mathematics, as once it has already done, the means of renewing its life and of developing itself anew. It is like the Giant Antaeus, who renewed, his strength by touching the earth."

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FOOTNOTES

1 The more general notion of _anharmonic ratio_, which includes the harmonic ratio as a special case, was also known to the ancients. While we have not found it necessary to make use of the anharmonic ratio in building up our theory, it is so frequently met with in treatises on geometry that some account of it should be given.

Consider any four points, _A_, _B_, _C_, _D_, on a line, and join them to any point _S_ not on that line. Then the triangles _ASB_, _GSD_, _ASD_, _CSB_, having all the same altitude, are to each other as their bases. Also, since the area of any triangle is one half the product of any two of its sides by the sine of the angle included between them, we have

[formula]

Now the fraction on the right would be unchanged if instead of the points _A_, _B_, _C_, _D_ we should take any other four points _A'_, _B'_, _C'_, _D'_ lying on any other line cutting across _SA_, _SB_, _SC_, _SD_. In other words, _the fraction on the left is unaltered in value if the points __A__, __B__, __C__, __D__ are replaced by any other four points perspective to them._ Again, the fraction on the left is unchanged if some other point were taken instead of _S_. In other words, _the fraction on the right is unaltered if we replace the four lines __SA__, __SB__, __SC__, __SD__ by any other four lines perspective to them._ The fraction on the left is called the _anharmonic ratio_ of the four points _A_, _B_, _C_, _D_; the fraction on the right is called the _anharmonic ratio_ of the four lines _SA_, _SB_, _SC_, _SD_. The anharmonic ratio of four points is sometimes written (_ABCD_), so that

[formula]

If we take the points in different order, the value of the anharmonic ratio will not necessarily remain the same. The twenty-four different ways of writing them will, however, give not more than six different values for the anharmonic ratio, for by writing out the fractions which define them we can find that _(ABCD) = (BADC) = (CDAB) = (DCBA)_. If we write _(ABCD) = a_, it is not difficult to show that the six values are

[formula]

The proof of this we leave to the student.

If _A_, _B_, _C_, _D_ are four harmonic points (see Fig. 6, p. *22), and a quadrilateral _KLMN_ is constructed such that _KL_ and _MN_ pass through _A_, _KN_ and _LM_ through _C_, _LN_ through _B_, and _KM_ through _D_, then, projecting _A_, _B_, _C_, _D_ from _L_ upon _KM_, we have _(ABCD) = (KOMD)_, where _O_ is the intersection of _KM_ with _LN_. But, projecting again the points _K_,

_O_, _M_, _D_ from _N_ back upon the line _AB_, we have _(KOMD) = (CBAD)_. From this we have

(ABCD) = (CBAD),

or

[formula]

whence _a = 0_ or _a = 2_. But it is easy to see that _a = 0_ implies that two of the four points coincide. For four harmonic points, therefore, the six values of the anharmonic ratio reduce to three, namely, 2, [formula], and -1. Incidentally we see that if an interchange of any two points in an anharmonic ratio does not change its value, then the four points are harmonic.

[Figure 49]

FIG. 49

Many theorems of projective geometry are succinctly stated in terms of anharmonic ratios. Thus, the _anharmonic ratio of any four elements of a form is equal to the anharmonic ratio of the corresponding four elements in any form projectively related to it. The anharmonic ratio of the lines joining any four fixed points on a conic to a variable fifthpoint on the conic is constant. The locus of points from which four points in a plane are seen along four rays of constant anharmonic ratio is a conic through the four points._ We leave these theorems for the student, who may also justify the following solution of the problem: _Given three points and a certain anharmonic ratio, to find a fourth point which shall have with the given three the given anharmonic ratio._ Let _A_, _B_, _D_ be the three given points (Fig. 49). On any convenient line through _A_ take two points _B’_ and _D’_ such that _AB’/AD’_ is equal to the given anharmonic ratio. Join _BB’_ and _DD’_ and let the two lines meet in _S_. Draw through _S_ a parallel to _AB’_. This line will meet _AB_ in the required point _C_.

2 Pappus, Mathematicae Collectiones, vii, 129.

3 J. Verneri, Libellus super vigintiduobus elementis conicis, etc. 1522.

4 Kepler, Ad Vitellionem paralipomena quibus astronomiae pars optica traditur. 1604.

5 Desargues, Bruillon-project d’une atteinte aux événements des rencontres d’un cône avec un plan. 1639. Edited and analyzed by Poudra, 1864.

6 The term ’pole’ was first introduced, in the sense in which we have used it, in 1810, by a French mathematician named Servois (Gergonne, _Annales des Mathéématiques_, I, 337), and the corresponding term ’polar’ by the editor, Gergonne, of this same journal three years later.

7 Euler, Introductio in analysin infinitorum, Appendix, cap. V. 1748.

8 Œuvres de Desargues, t. II, 132.

9 Œuvres de Desargues, t. II, 370.

10 Œuvres de Descartes, t. II, 499.

11 Œuvres de Pascal, par Brunsehvig et Boutroux, t. I, 252.

12 Chasles, Histoire de la Géométrie, 70.

13 Œuvres de Desargues, t. I, 231.

14 See Ball, History of Mathematics, French edition, t. II, 233.

15 Newton, Principia, lib. i, lemma XXI.

16 Maclaurin, Philosophical Transactions of the Royal Society of London, 1735.

17 Monge, Géométrie Descriptive. 1800.

18 Poncelet, Traité des Propriétés Projectives des Figures. 1822. (See p. 357, Vol. II, of the edition of 1866.)

19 Gergonne, _Annales de Mathématiques, XVI, 209. 1826._

20 Steiner, Systematische Ehtwickelung der Abhängigkeit geometrischer Gestalten von einander. 1832.

21 Von Staudt, Geometrie der Lage. 1847.

22 Reye, Geometrie der Lage. Translated by Holgate, 1897.

23 Ball, loc. cit. p. 261.